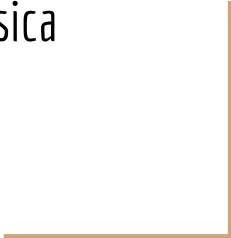




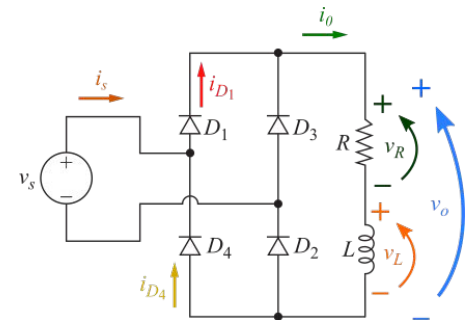
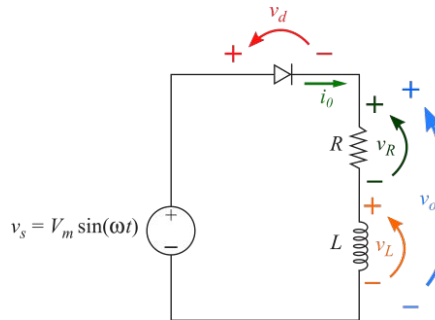
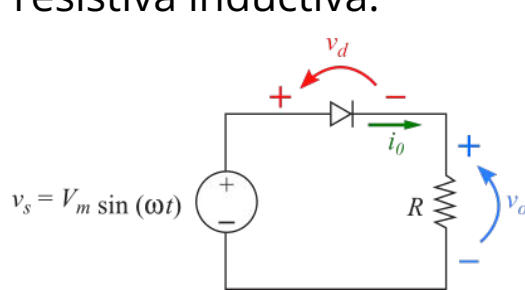
Conversión CA/CC

Rectificación Monofásica



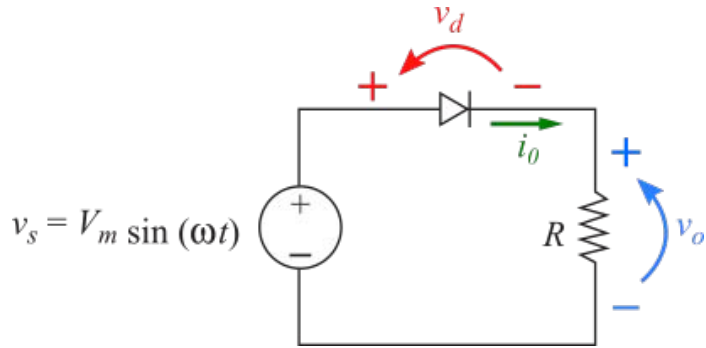
Rectificador a diodo

- El rectificador con diodos es la estructura típica utilizada en numerosas fuentes de alimentación, tanto reguladas como no reguladas.
- También se lo denomina rectificador **no controlado**.
 - La tensión media de salida depende de la tensión de entrada y **de la carga**.
 - No tengo ninguna variable de control para ajustar el valor medio de tensión.
- Veremos el rectificador de media onda y de onda completa, con carga resistiva y resistiva inductiva.



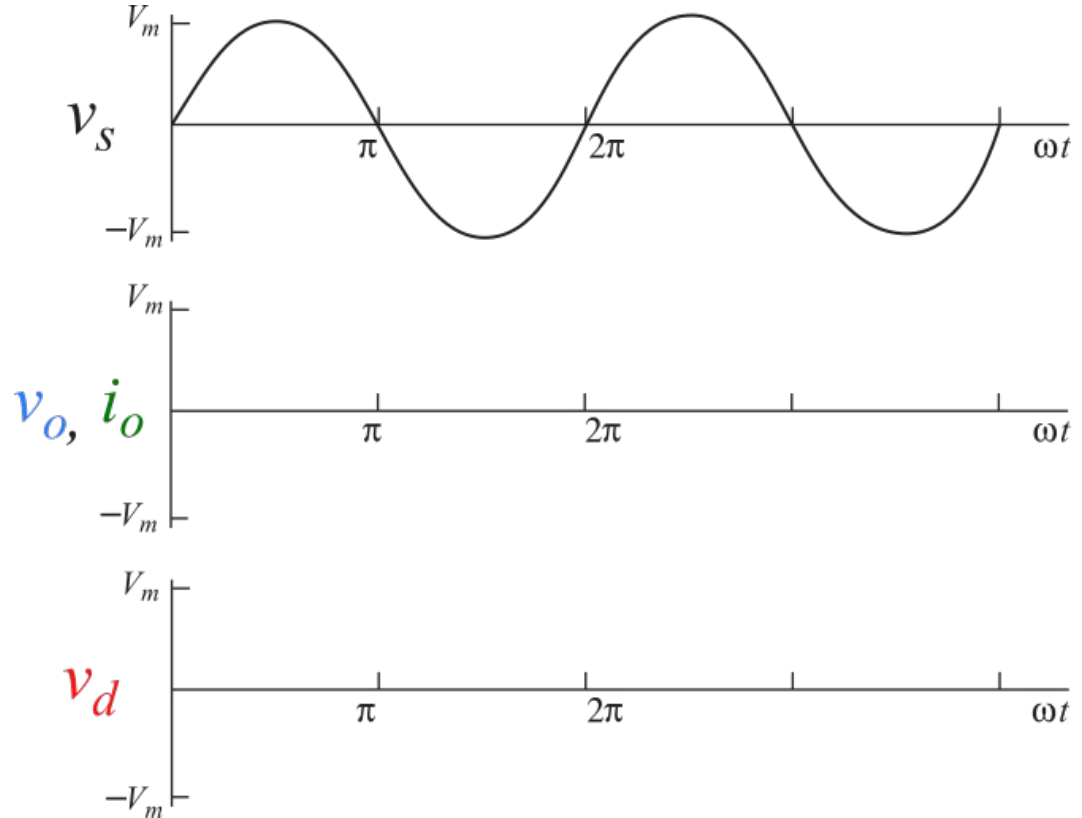
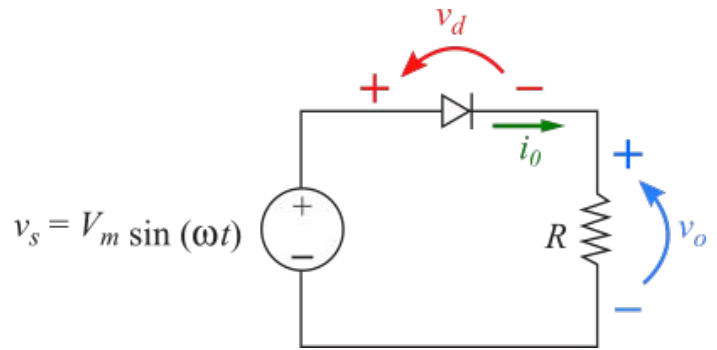
Rectificador de media onda

- Carga resistiva



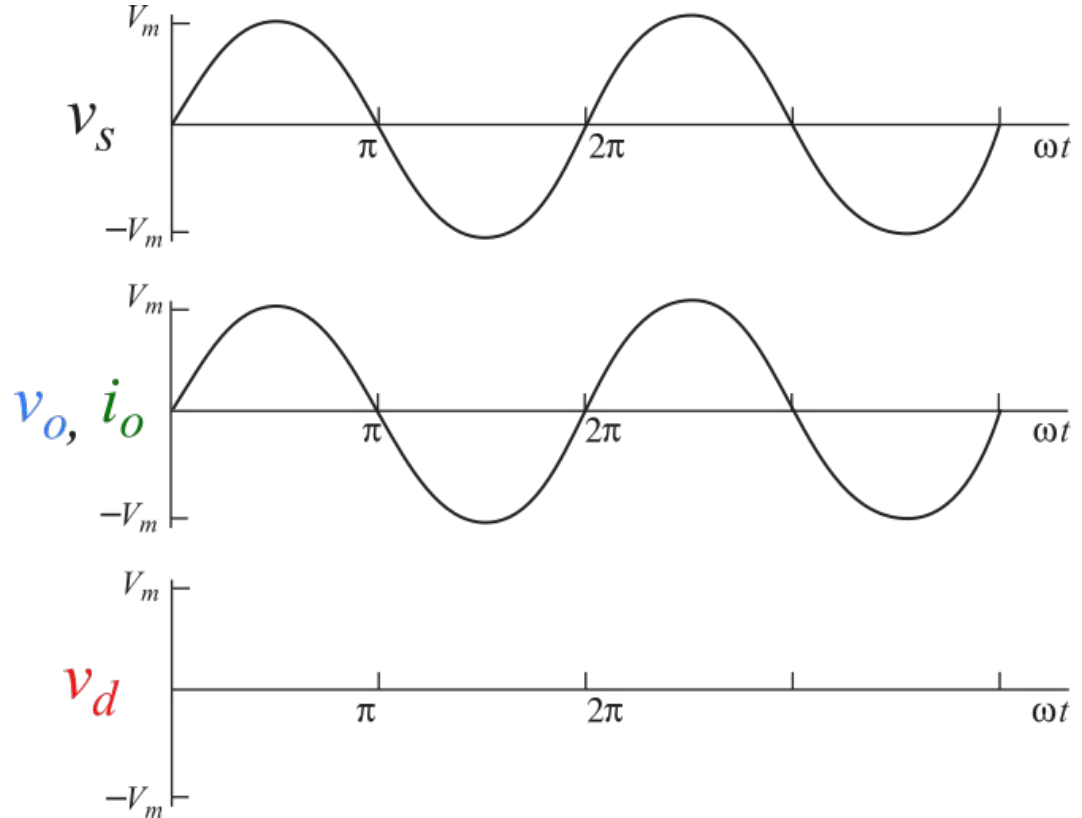
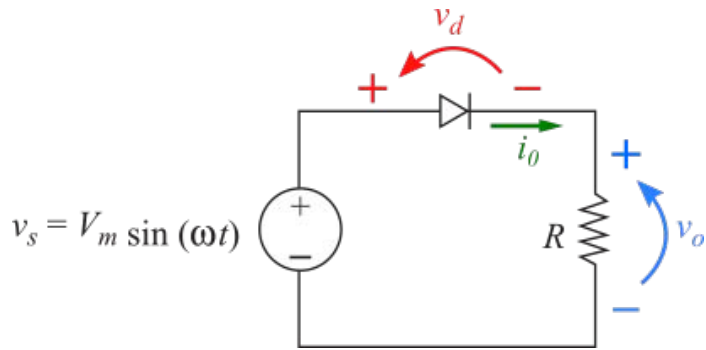
Rectificador de media onda

- Carga resistiva



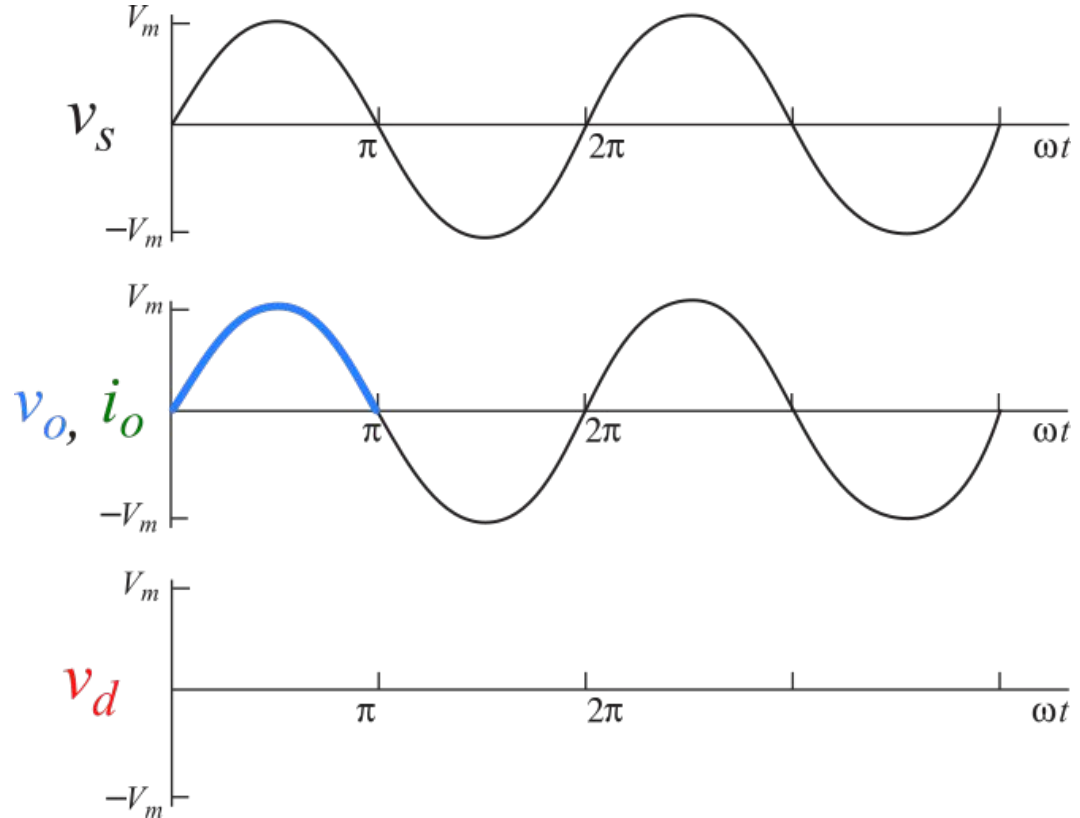
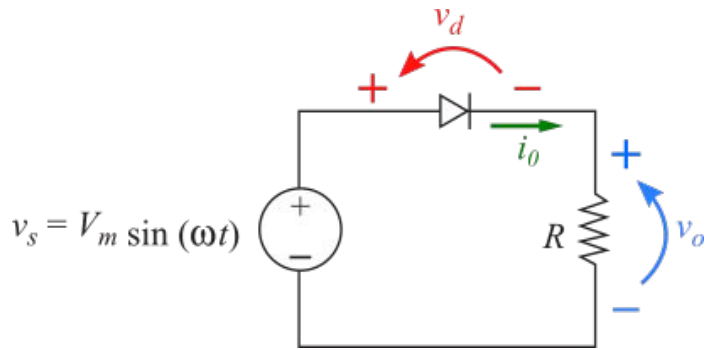
Rectificador de media onda

- Carga resistiva



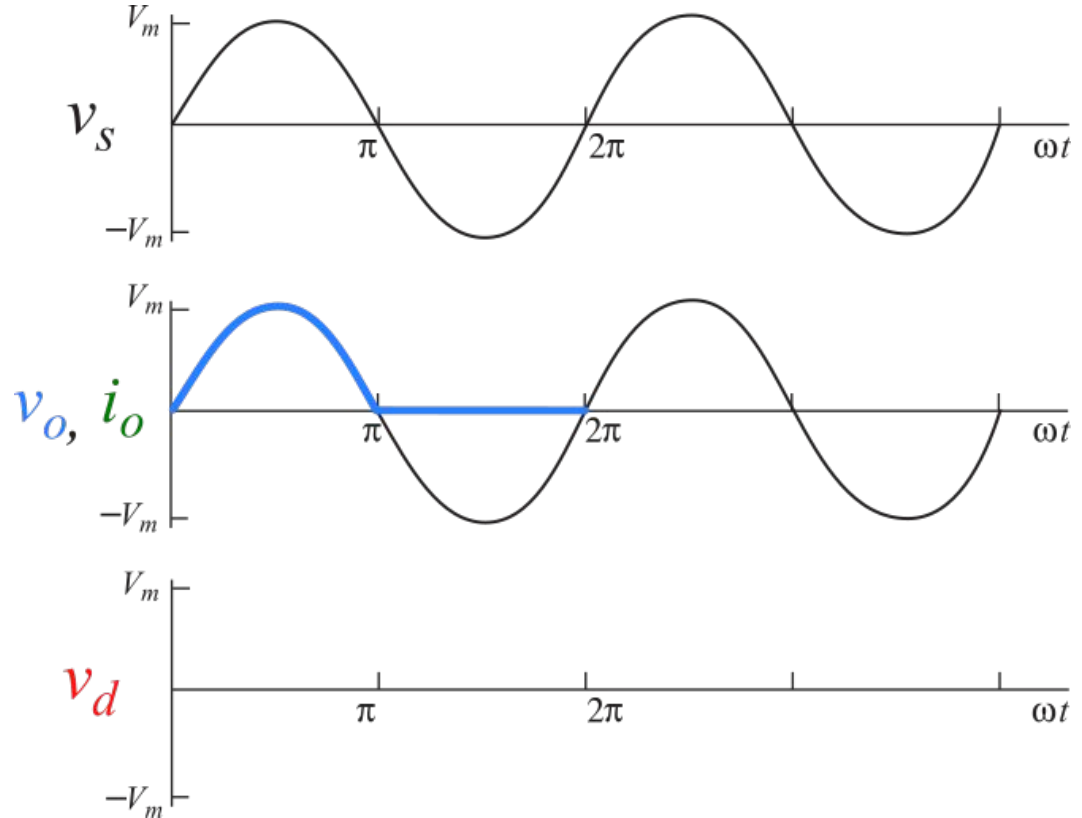
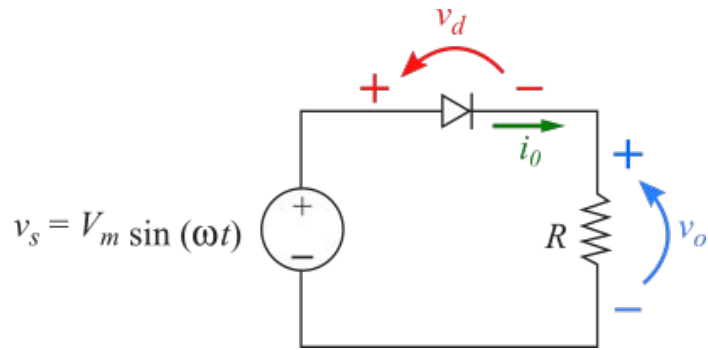
Rectificador de media onda

- Carga resistiva



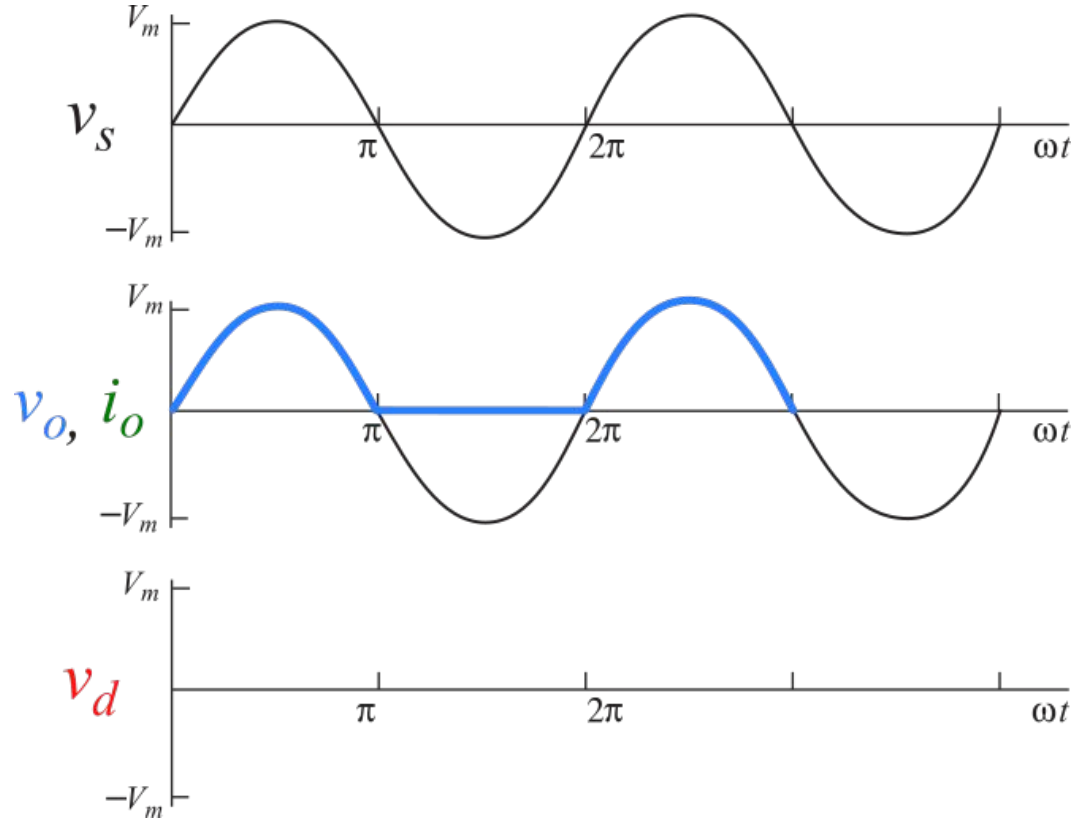
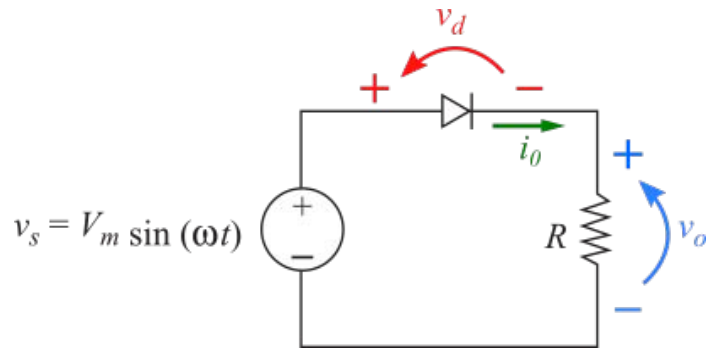
Rectificador de media onda

- Carga resistiva



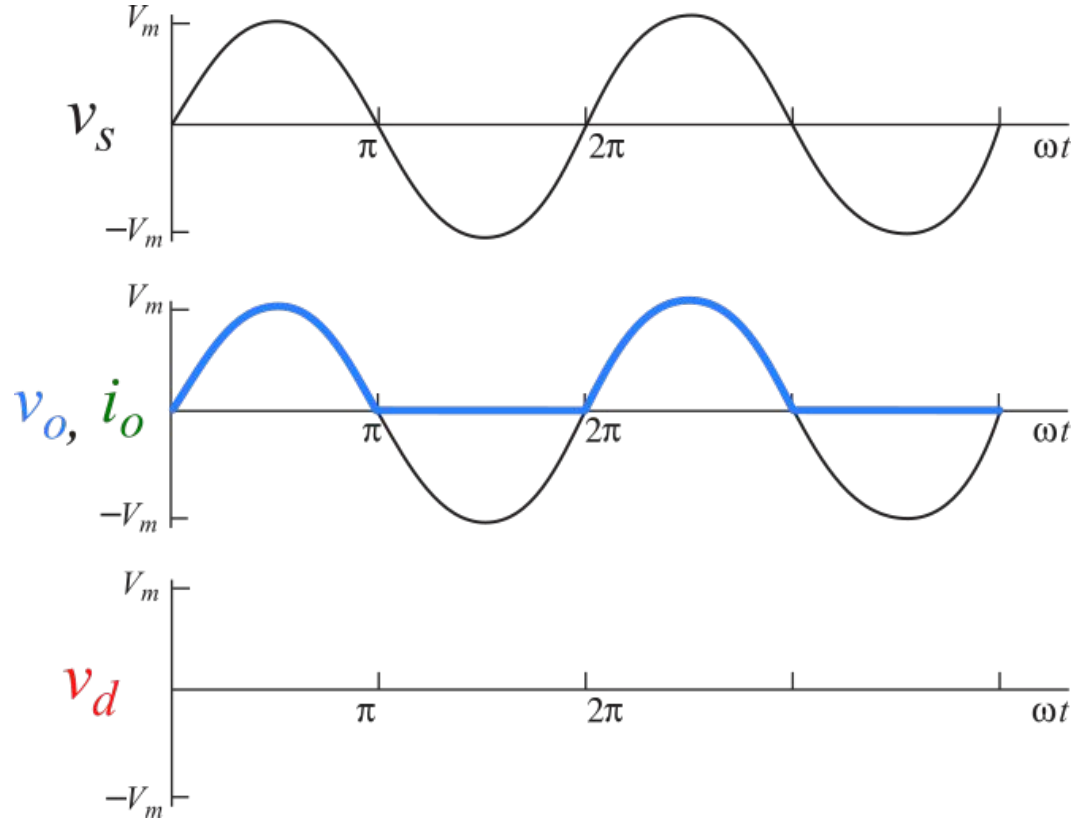
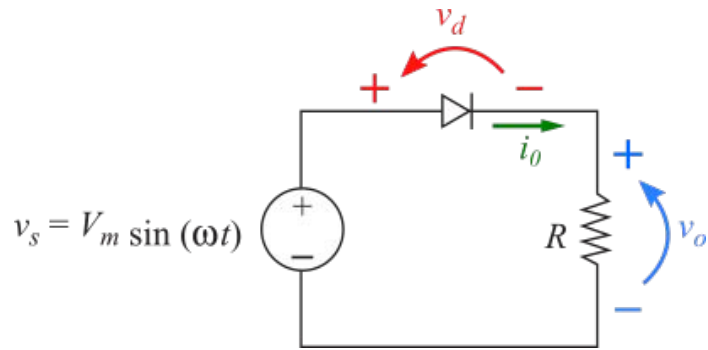
Rectificador de media onda

- Carga resistiva



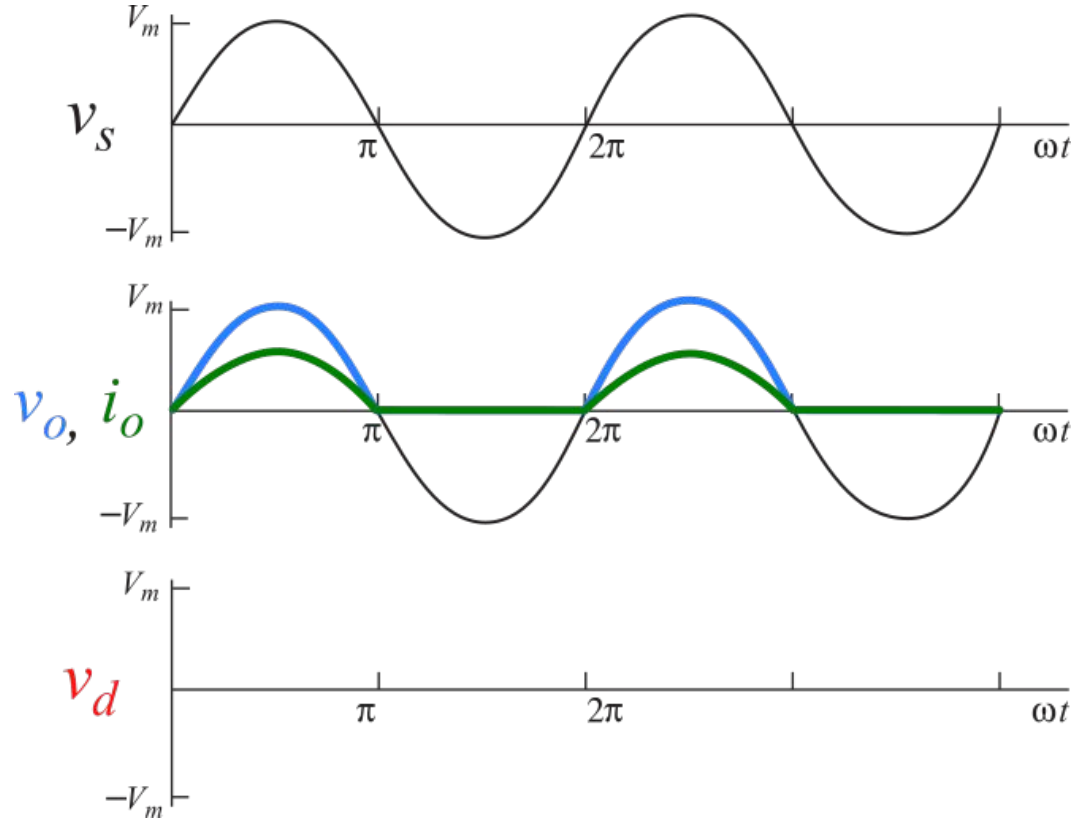
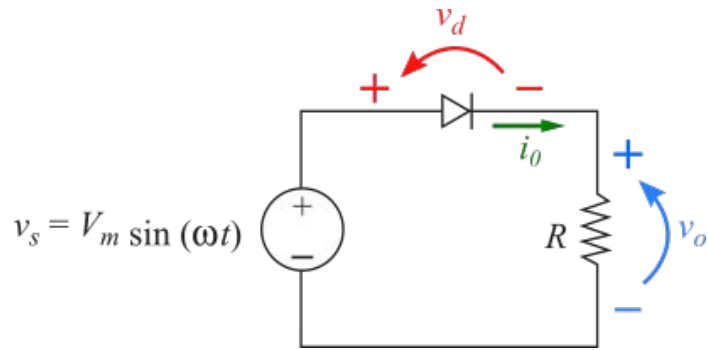
Rectificador de media onda

- Carga resistiva



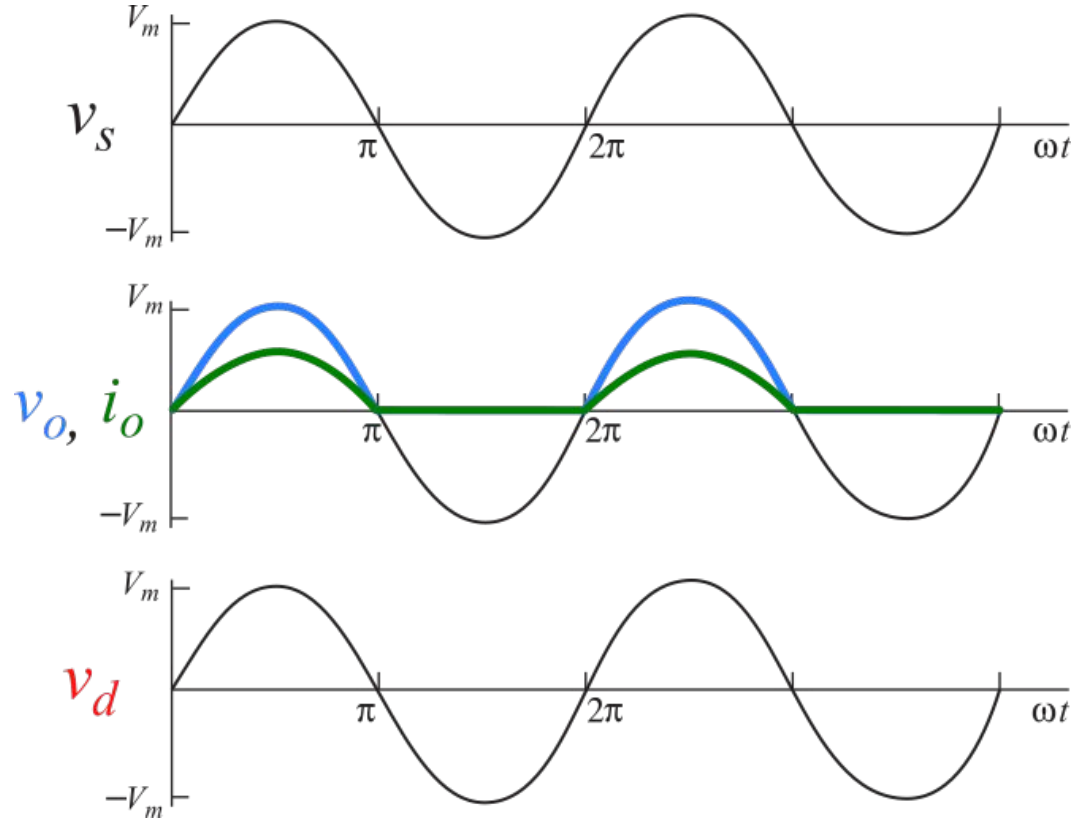
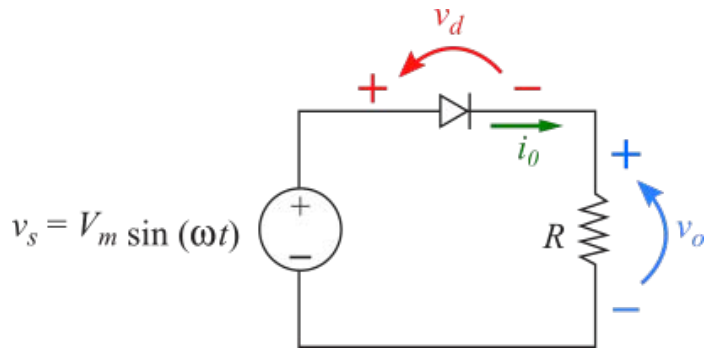
Rectificador de media onda

- Carga resistiva



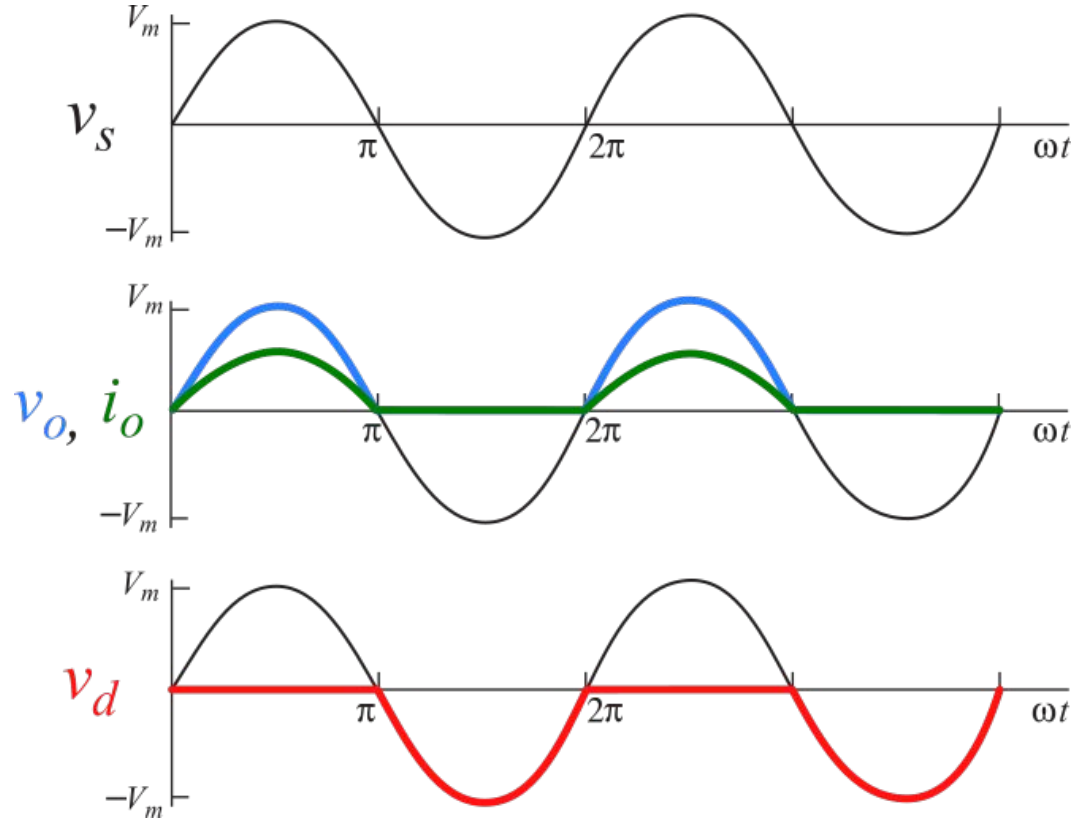
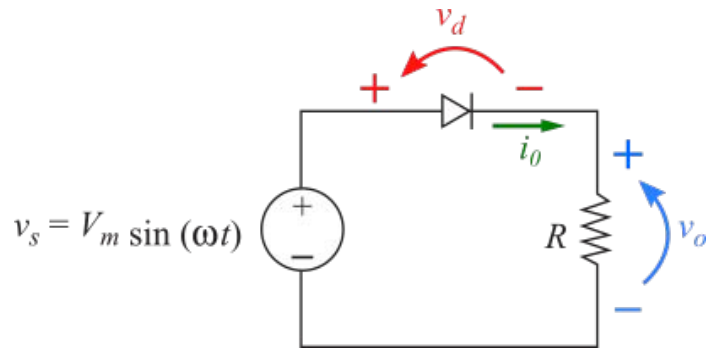
Rectificador de media onda

- Carga resistiva



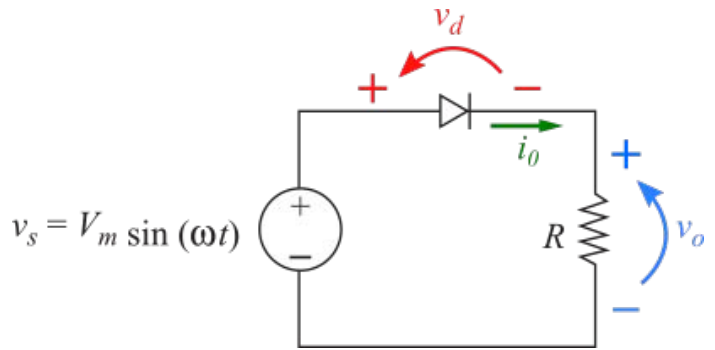
Rectificador de media onda

- Carga resistiva

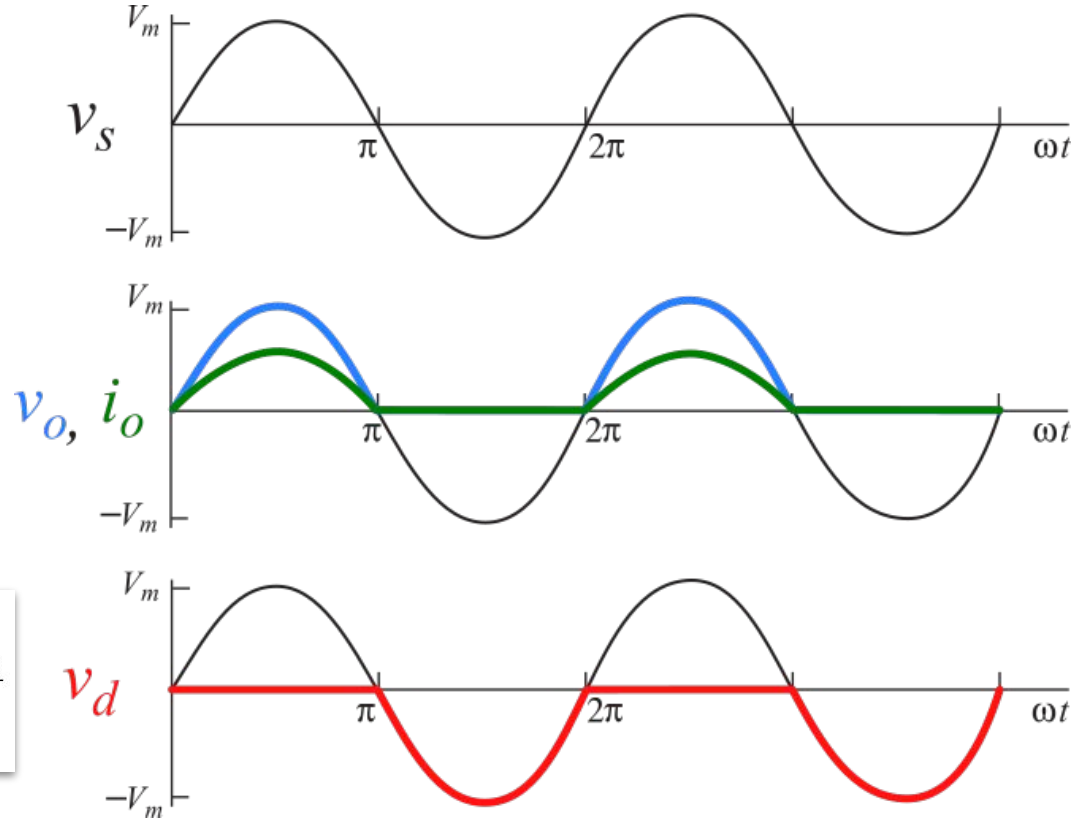


Rectificador de media onda

- Carga resistiva

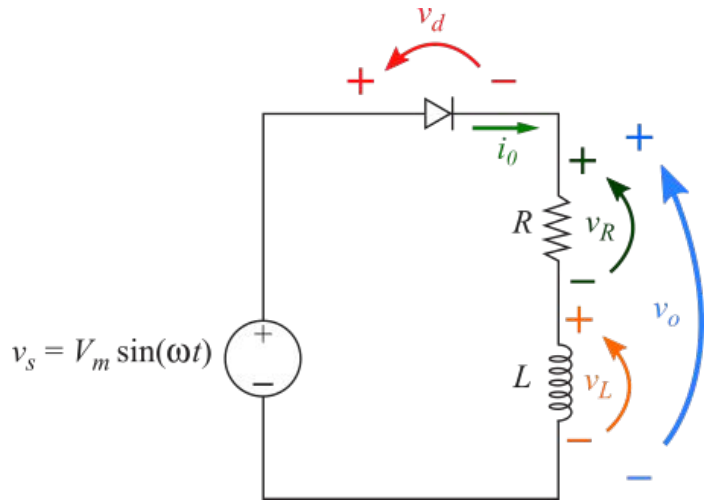


$$V_o = V_{\text{avg}} = \frac{1}{2\pi} \int_0^{\pi} V_m \sin(\omega t) d(\omega t) = \frac{V_m}{\pi}$$



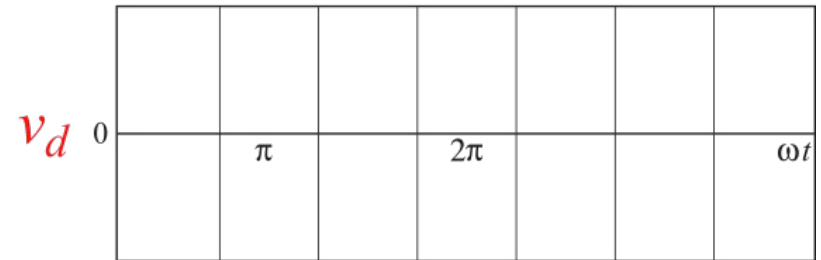
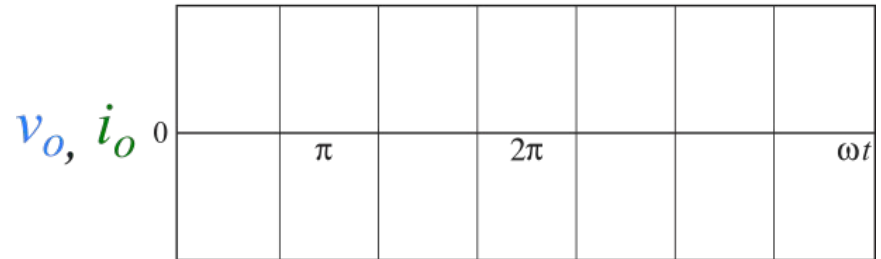
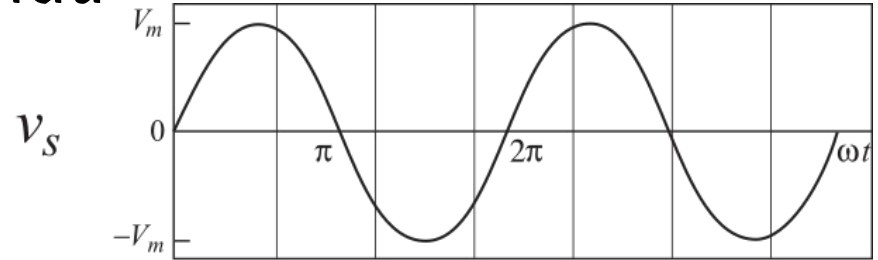
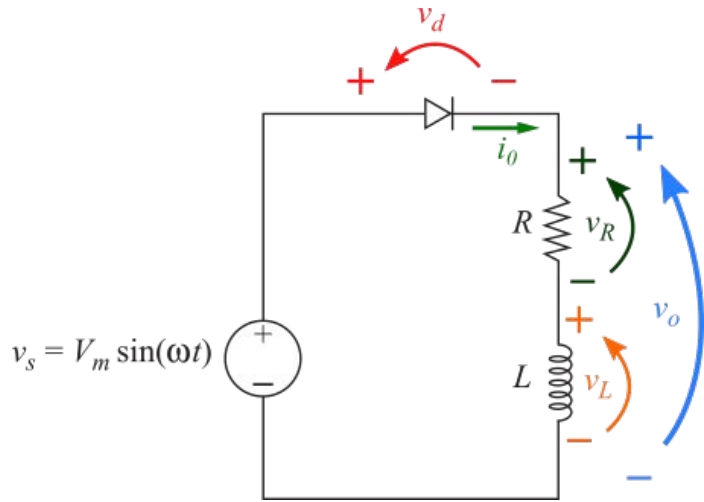
Rectificador de media onda

- Carga resistiva-inductiva



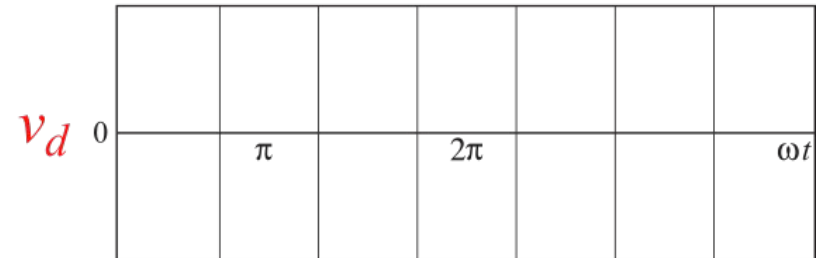
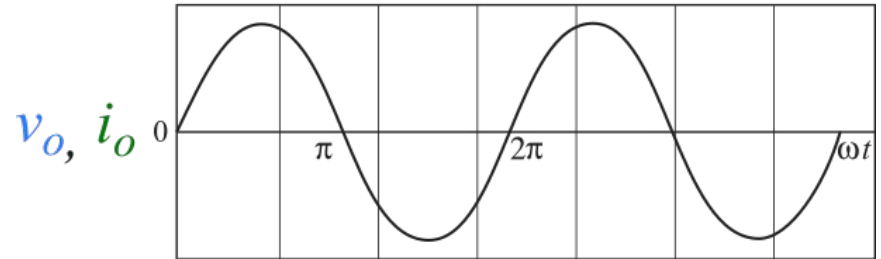
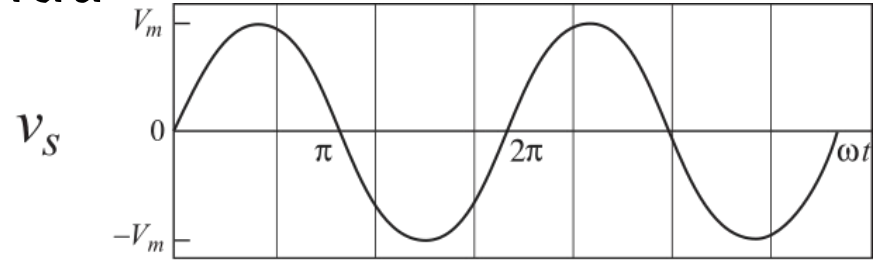
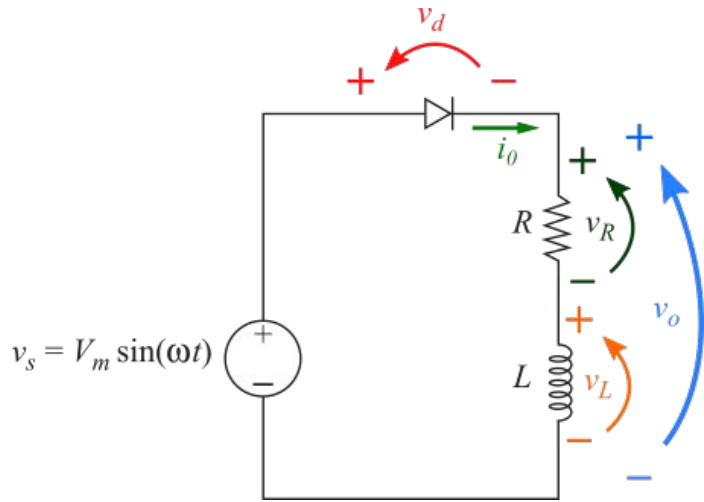
Rectificador de media onda

- Carga resistiva-inductiva



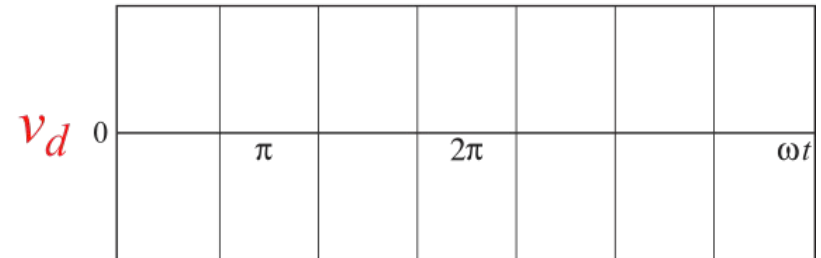
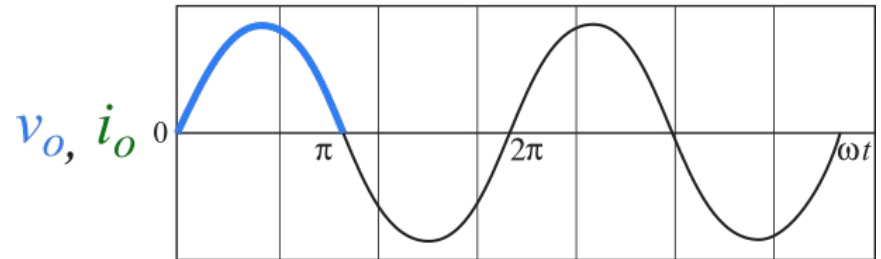
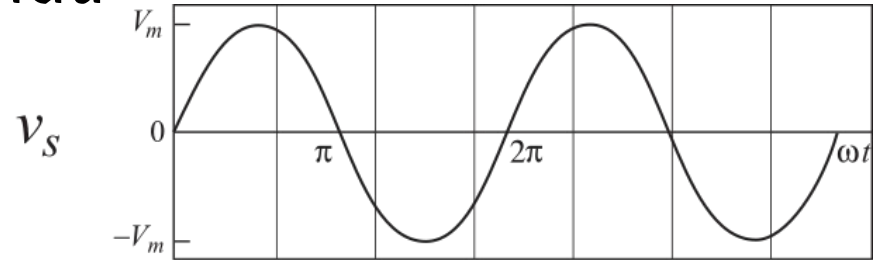
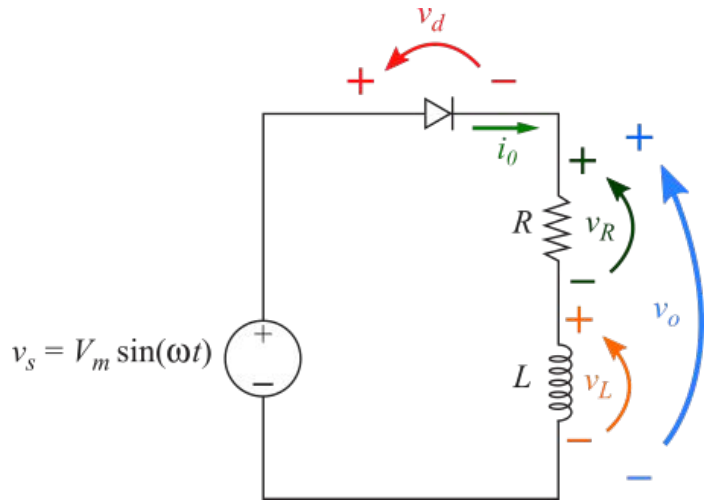
Rectificador de media onda

- Carga resistiva-inductiva



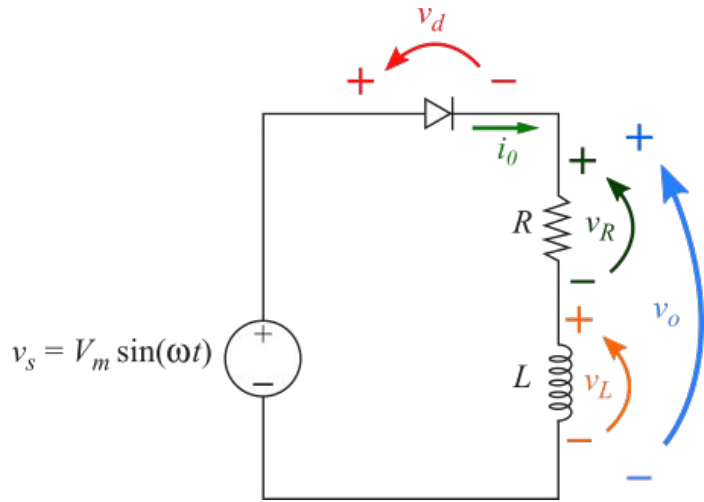
Rectificador de media onda

- Carga resistiva-inductiva

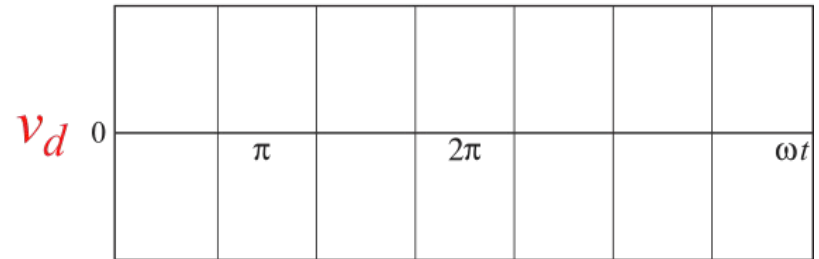
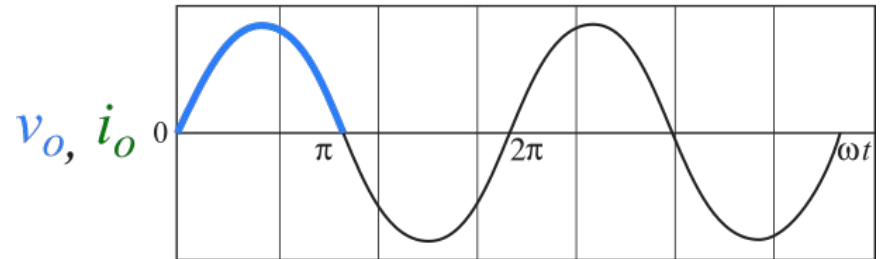
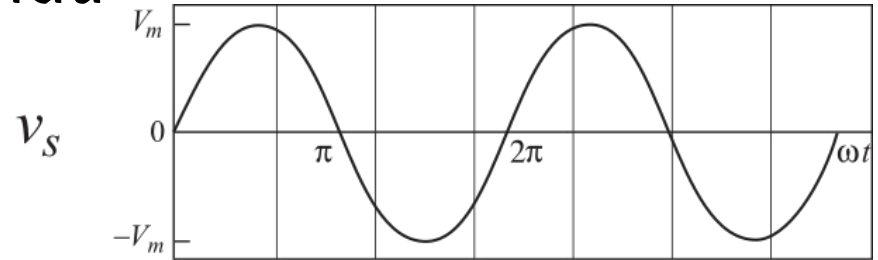


Rectificador de media onda

- Carga resistiva-inductiva



$$V_m \sin(\omega t) = Ri(t) + L \frac{di(t)}{dt}$$



Rectificador de media onda

- Carga resistiva-inductiva

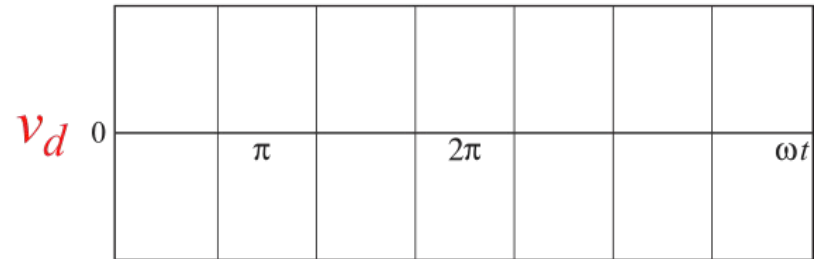
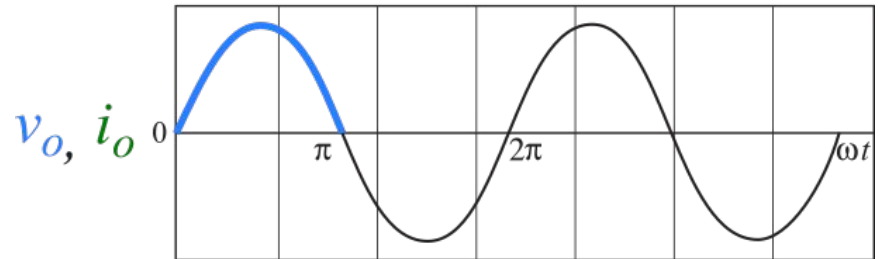
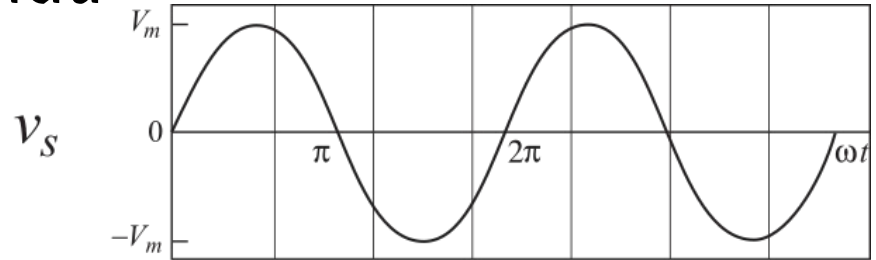
$$V_m \sin(\omega t) = Ri(t) + L \frac{di(t)}{dt}$$

$$i(t) = i_f(t) + i_n(t)$$

$$\rightarrow i_f(t) = \frac{V_m}{Z} \sin(\omega t - \theta)$$

$$Ri(t) + L \frac{di(t)}{dt} = 0$$

$$\rightarrow i_n(t) = Ae^{-t/\tau}$$



Rectificador de media onda

- Carga resistiva-inductiva

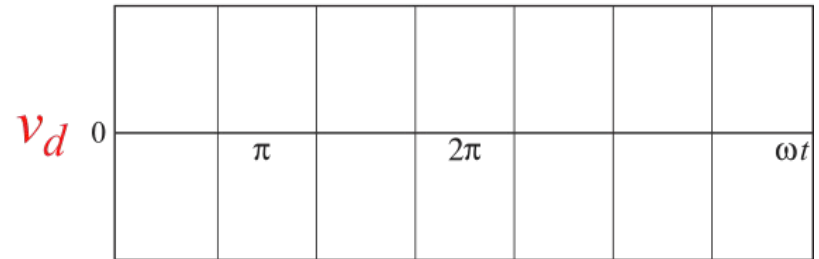
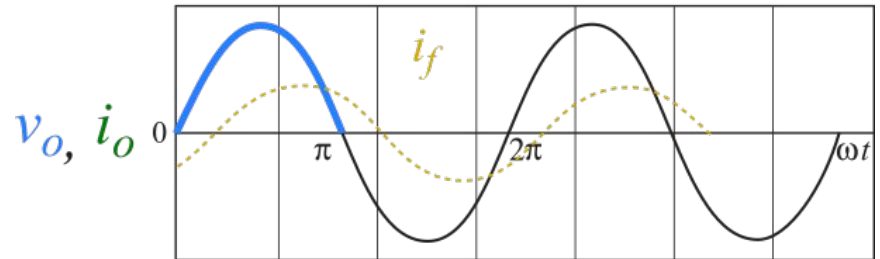
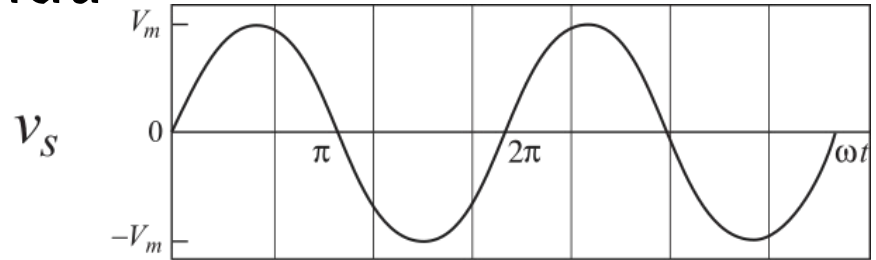
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Rectificador de media onda

- Carga resistiva-inductiva

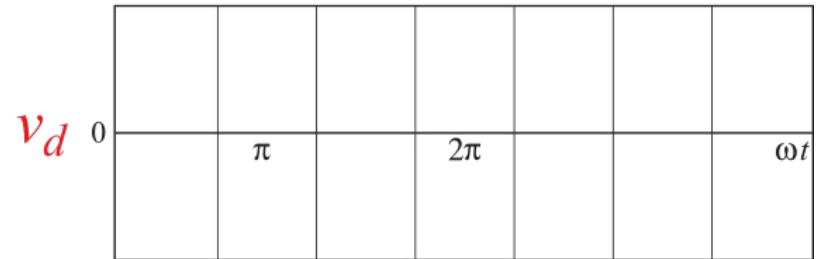
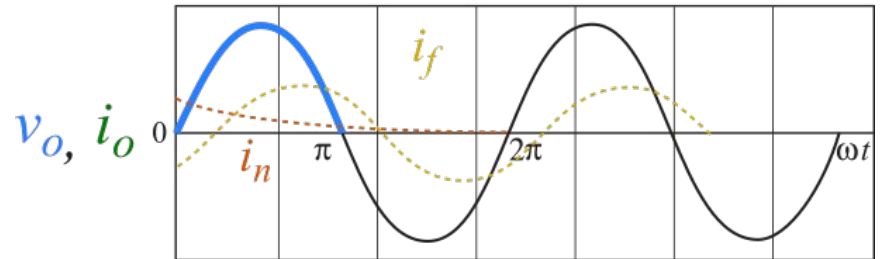
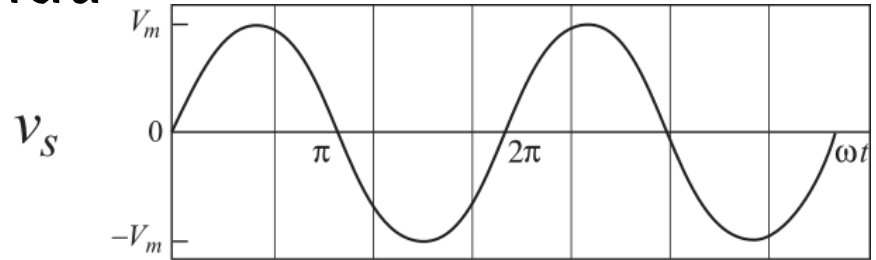
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Rectificador de media onda

- Carga resistiva-inductiva

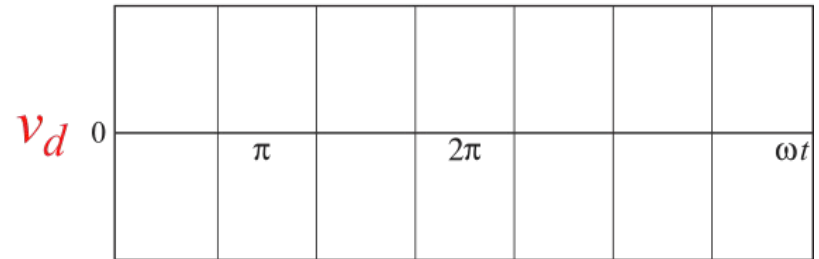
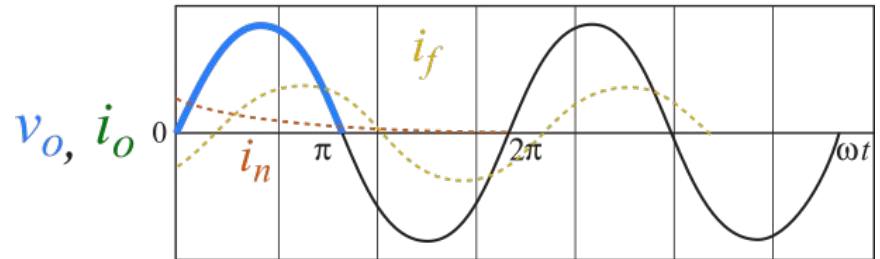
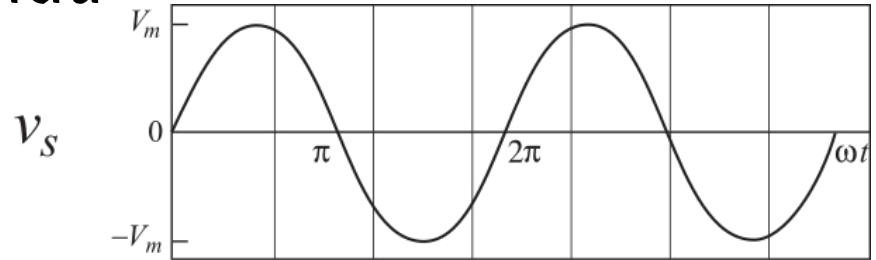
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$$A = -\frac{V_m}{Z} \sin(-\theta) = \frac{V_m}{Z} \sin \theta$$



Rectificador de media onda

- Carga resistiva-inductiva

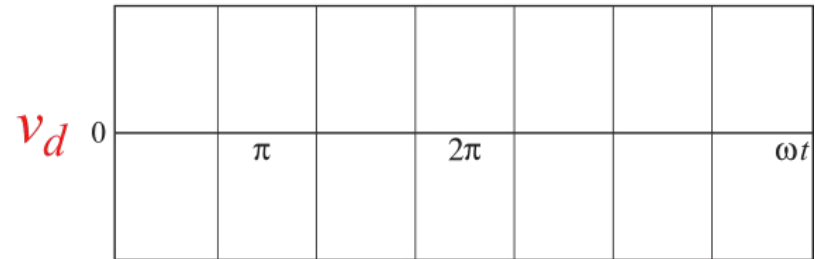
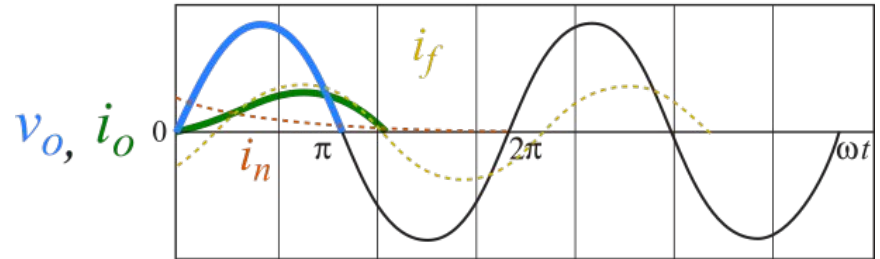
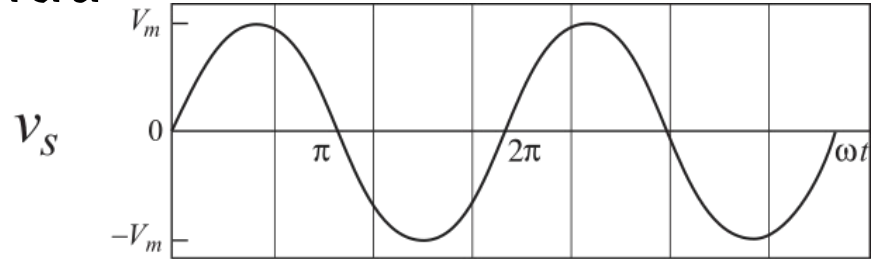
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Rectificador de media onda

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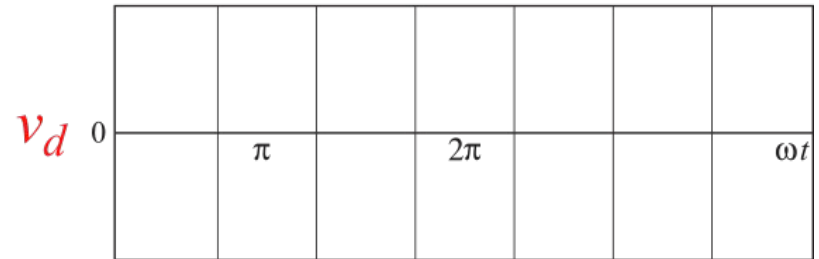
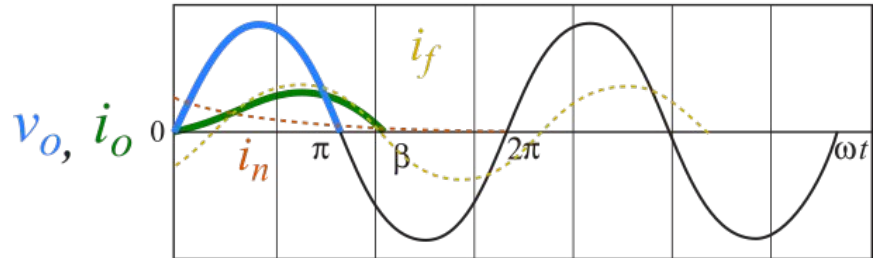
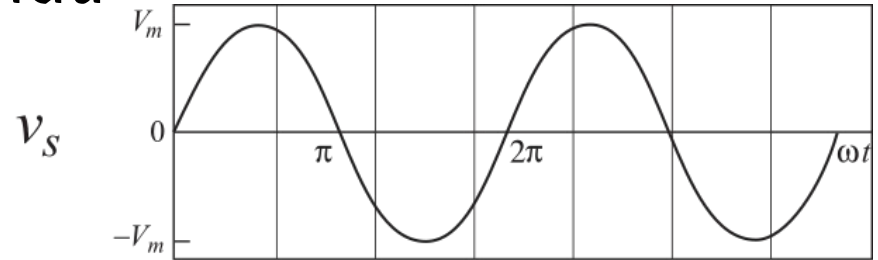
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Rectificador de media onda

- Carga resistiva-inductiva

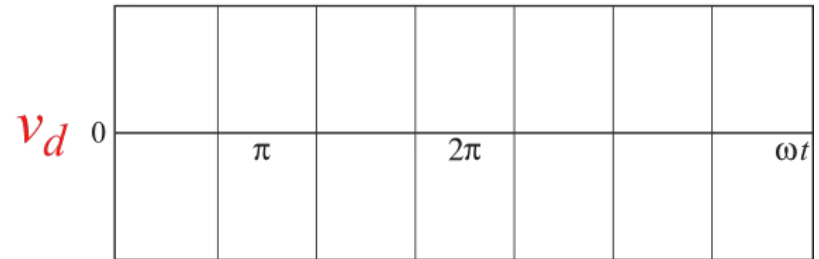
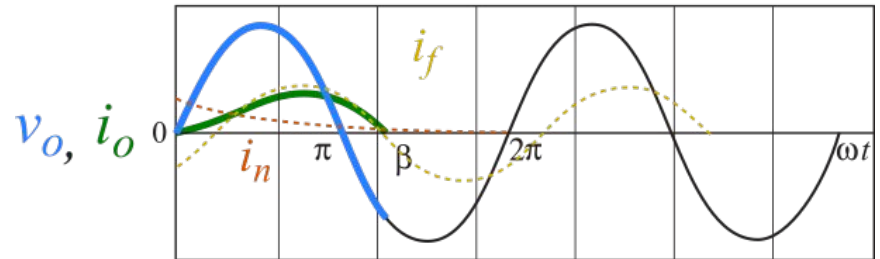
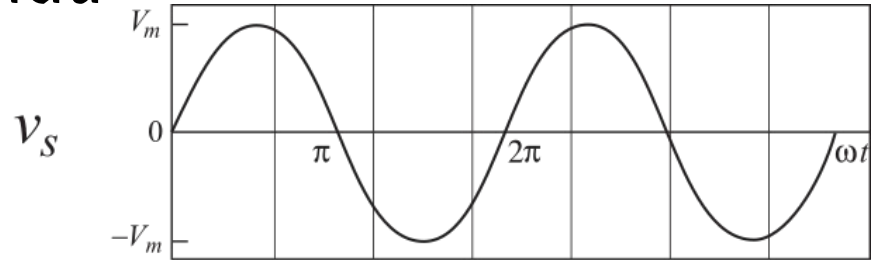
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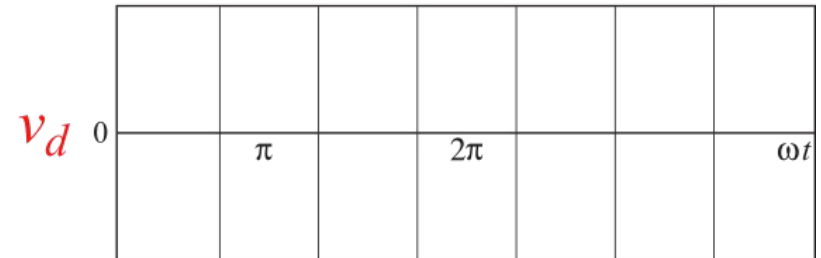
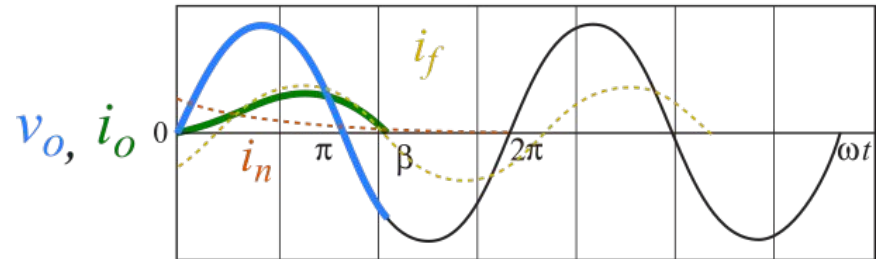
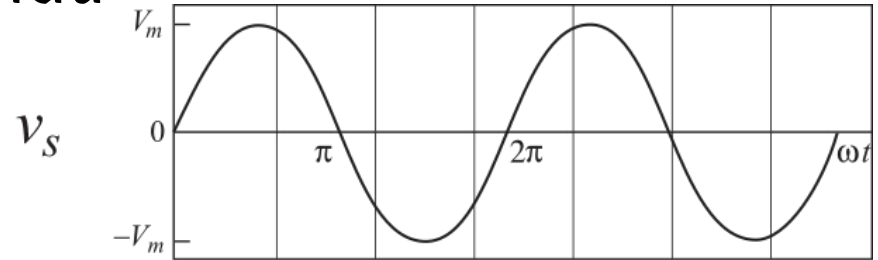
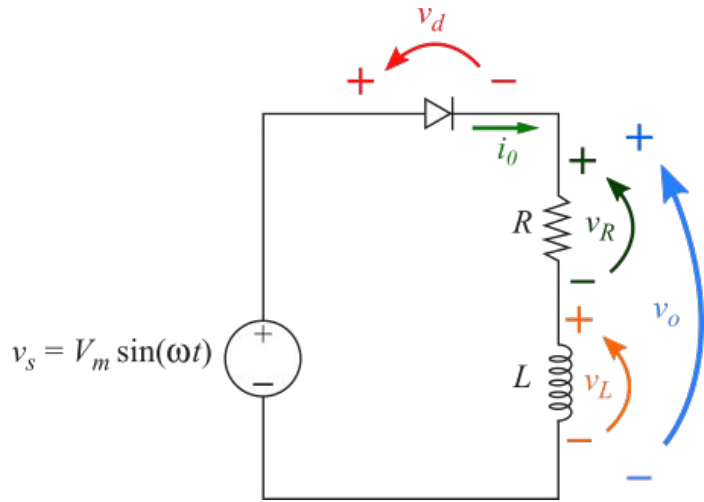
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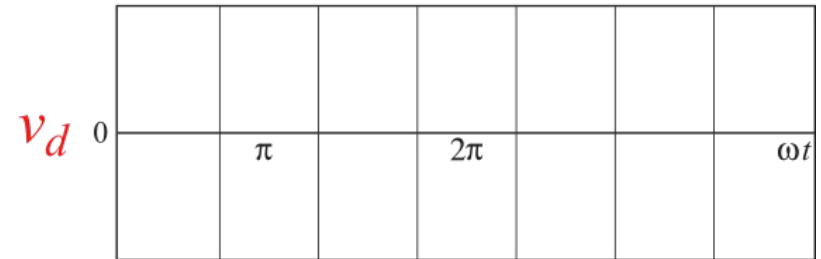
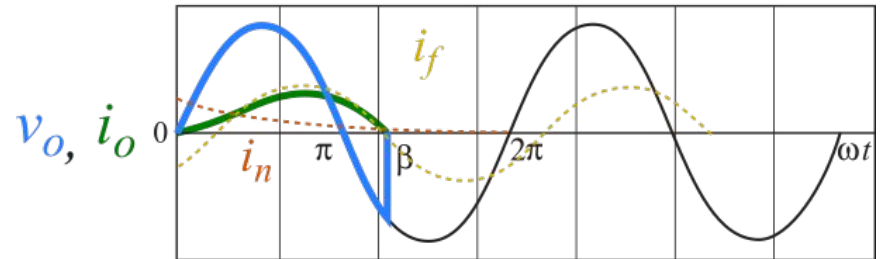
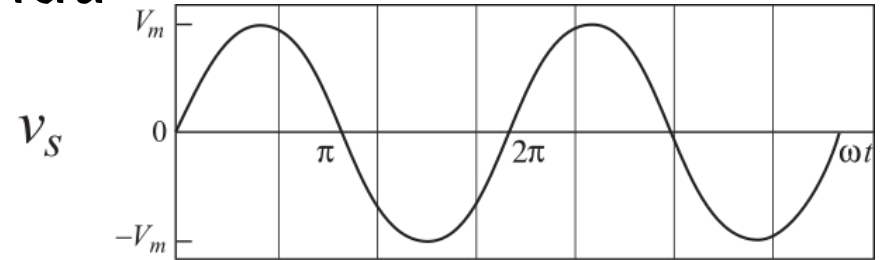
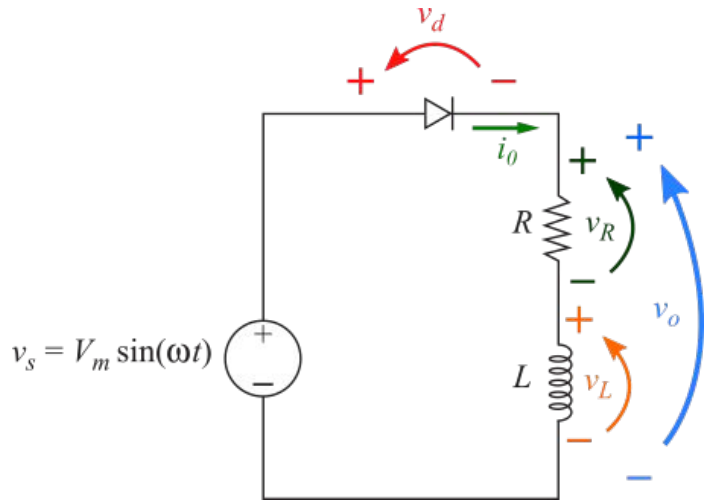
Rectificador de media onda

- Carga resistiva-inductiva



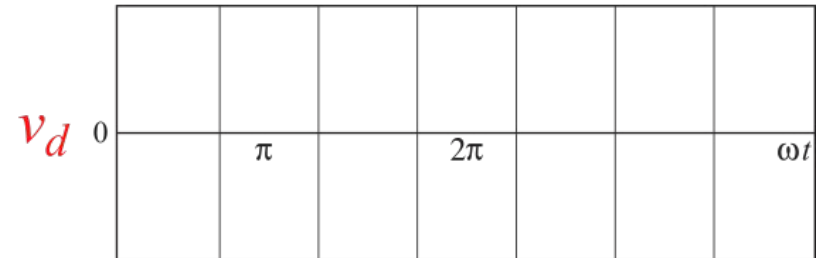
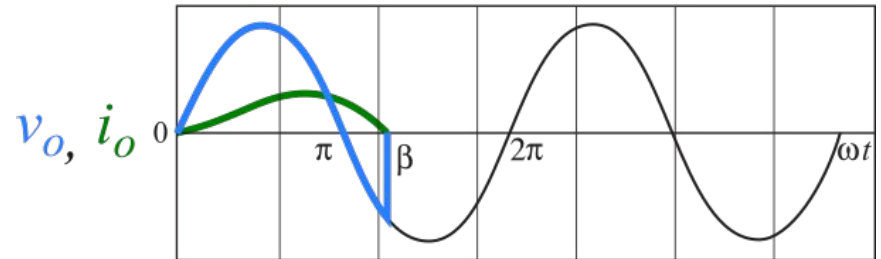
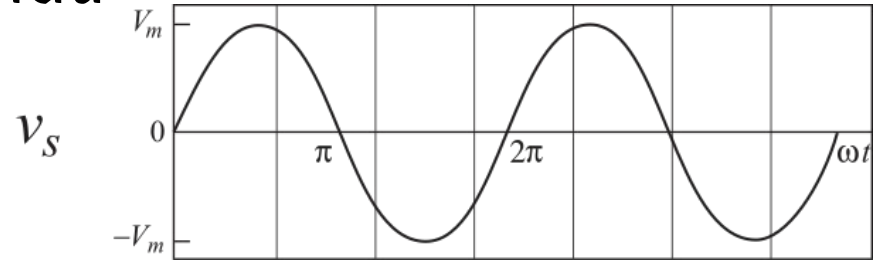
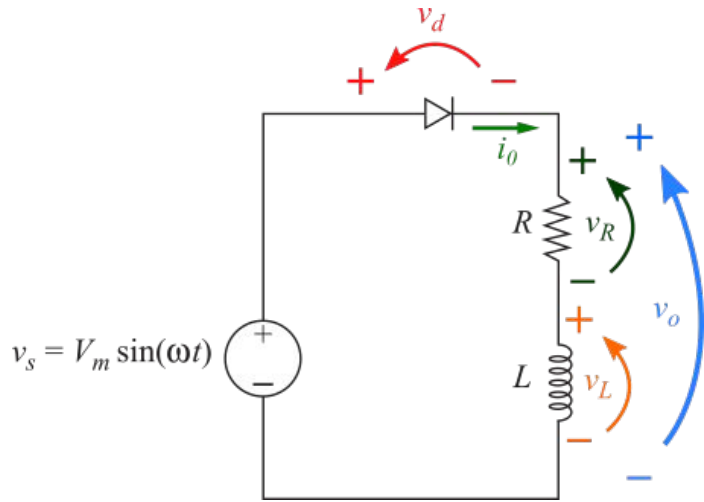
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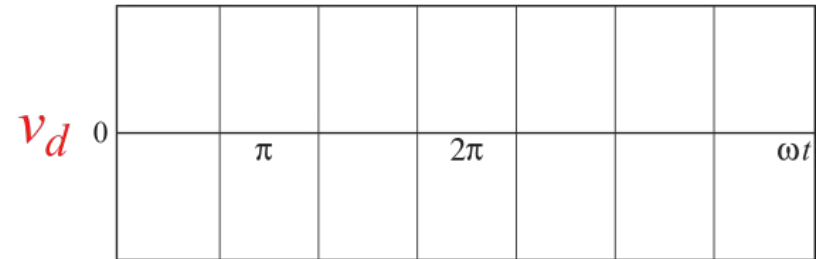
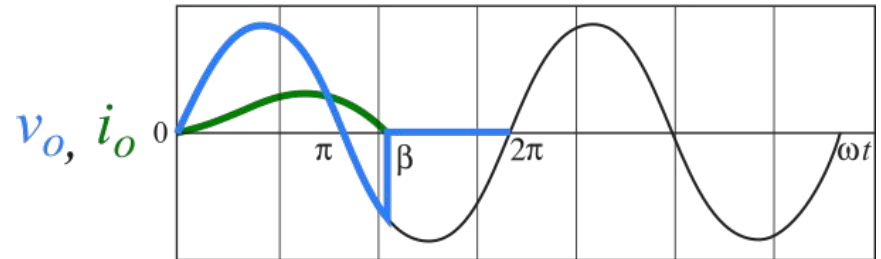
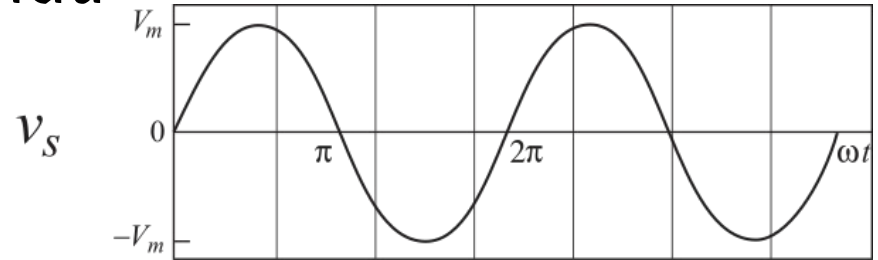
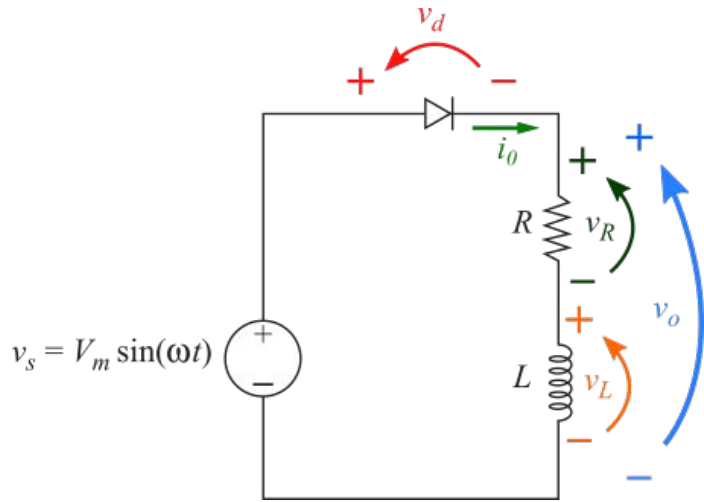
Rectificador de media onda

- Carga resistiva-inductiva



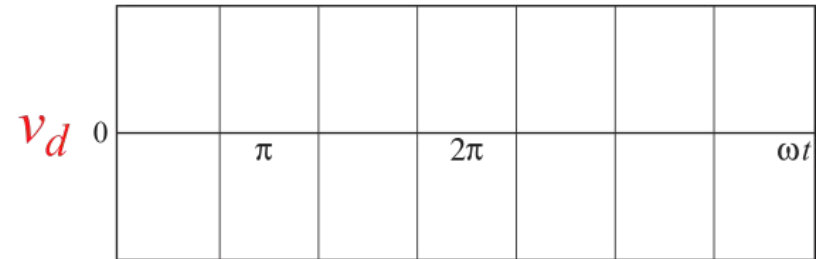
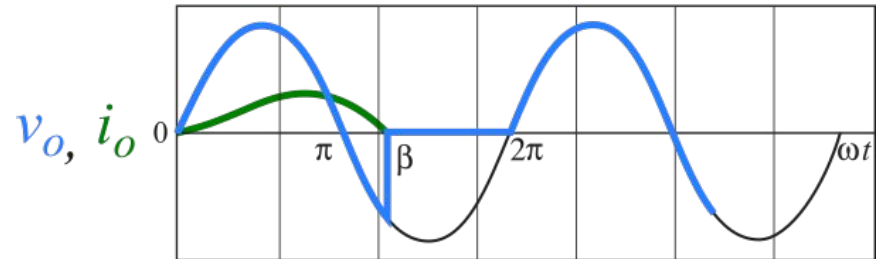
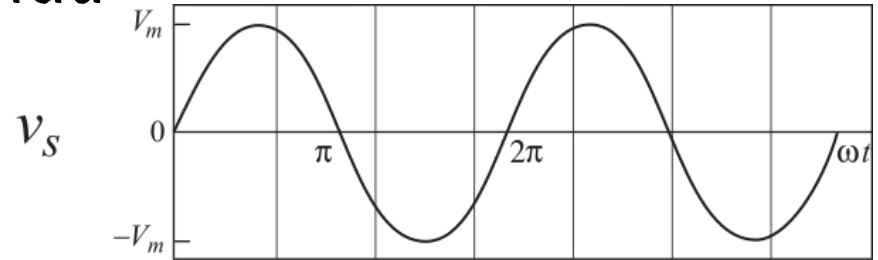
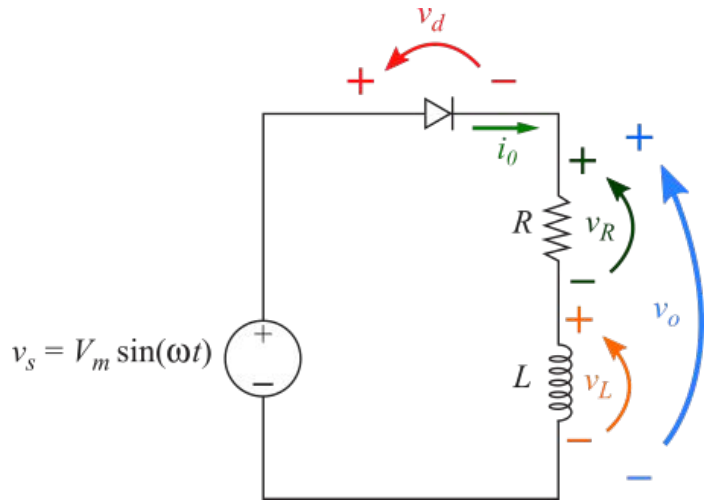
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- Carga resistiva-inductiva



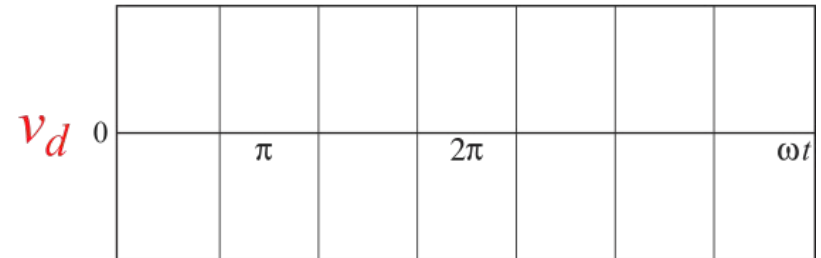
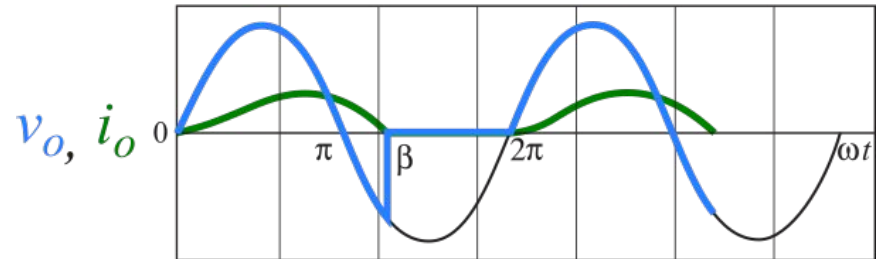
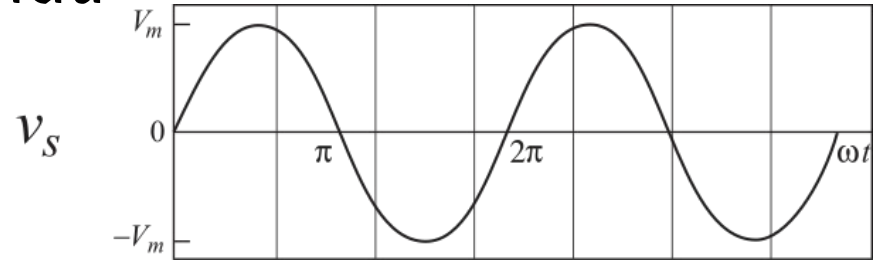
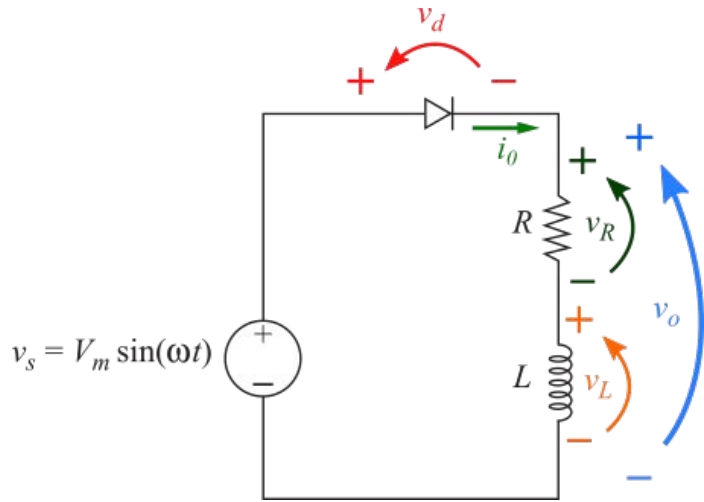
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- Carga resistiva-inductiva



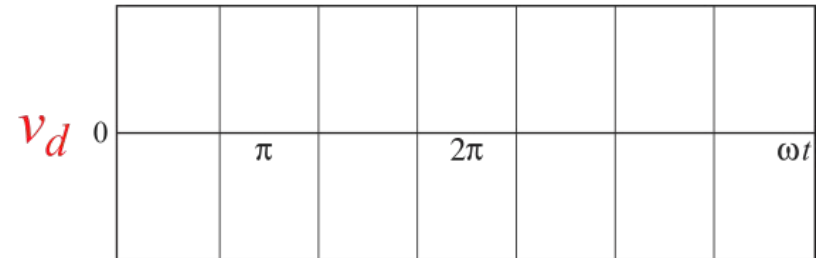
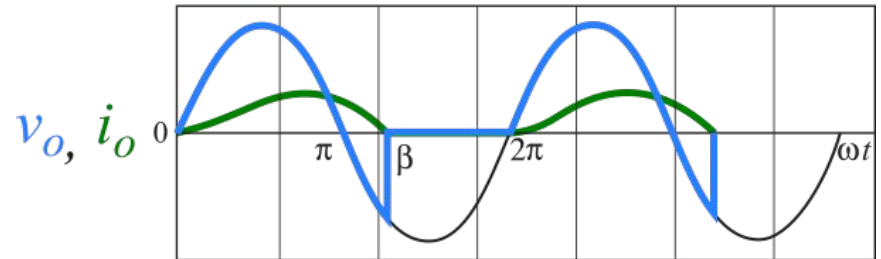
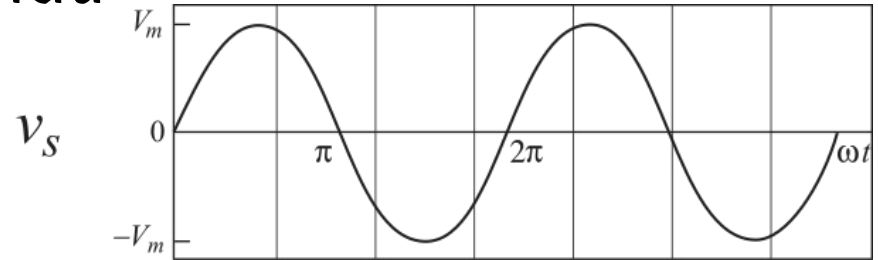
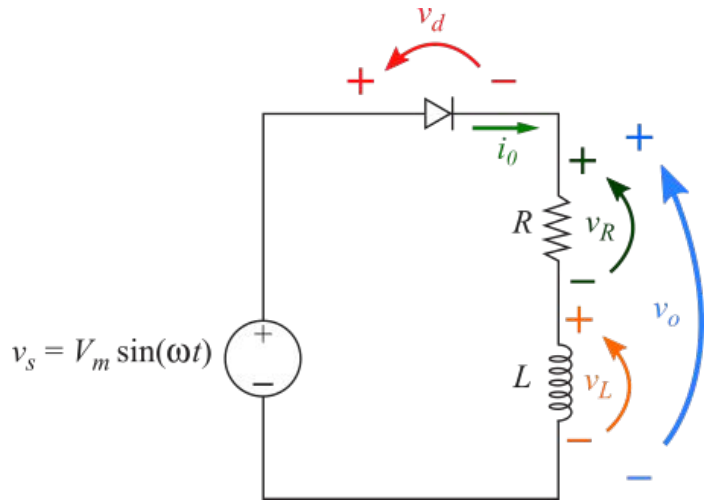
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- Carga resistiva-inductiva



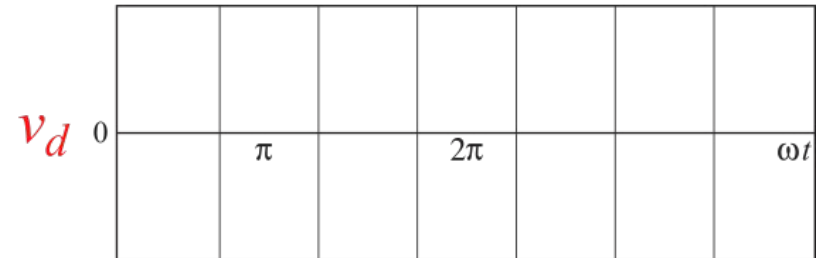
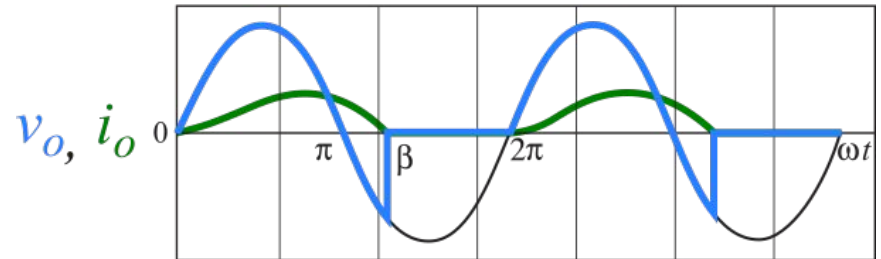
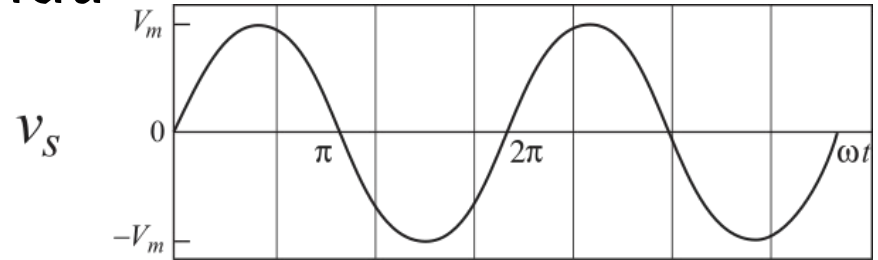
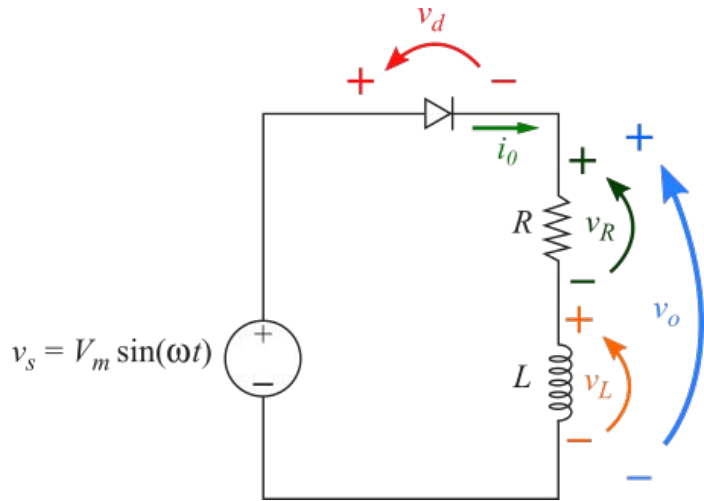
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- Carga resistiva-inductiva



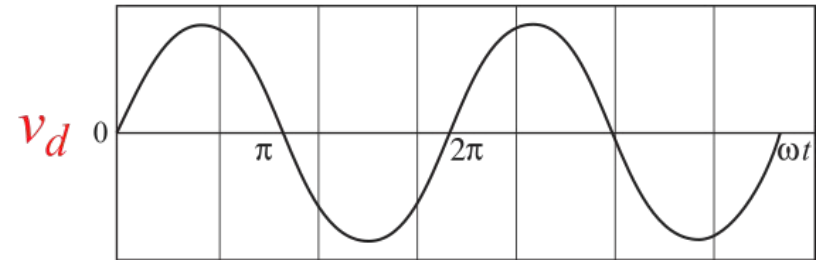
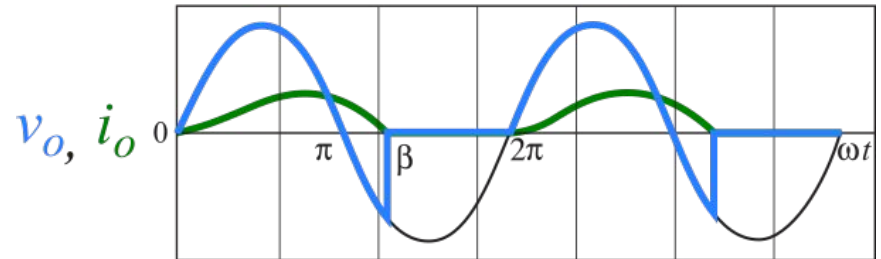
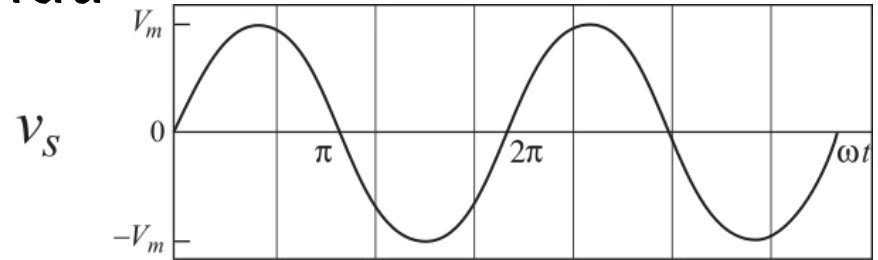
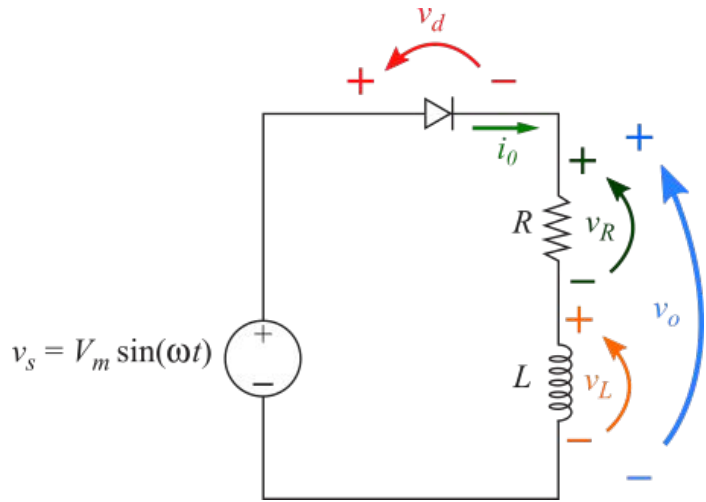
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- Carga resistiva-inductiva



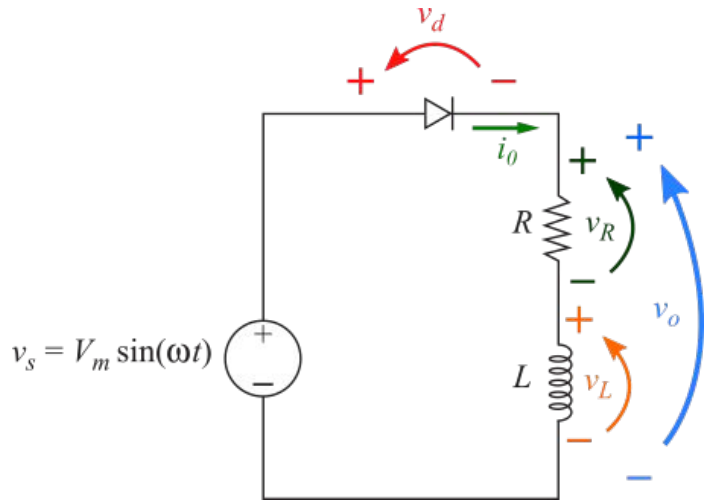
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- Carga resistiva-inductiva

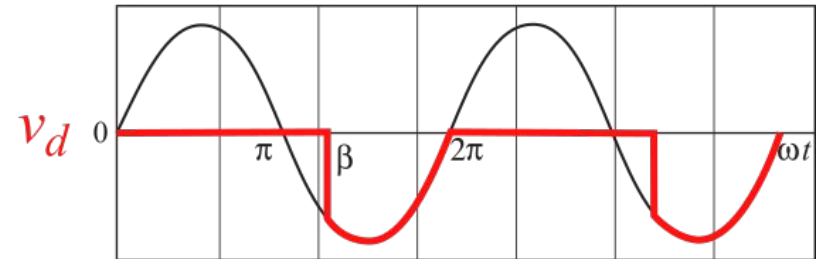
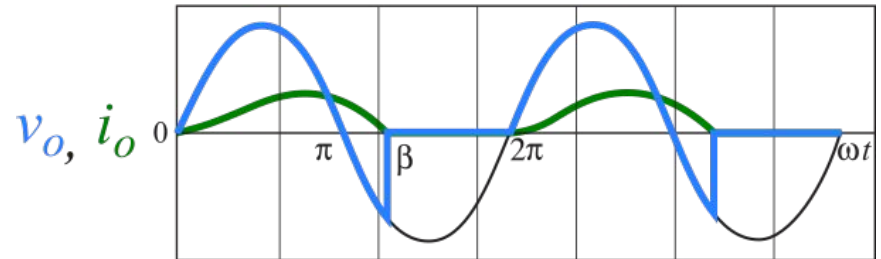
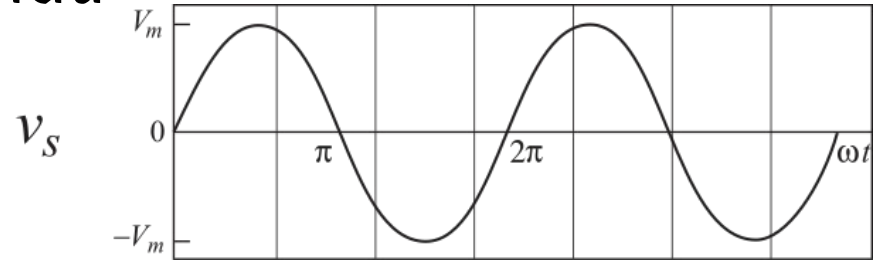


Rectificador de media onda

- Carga resistiva-inductiva

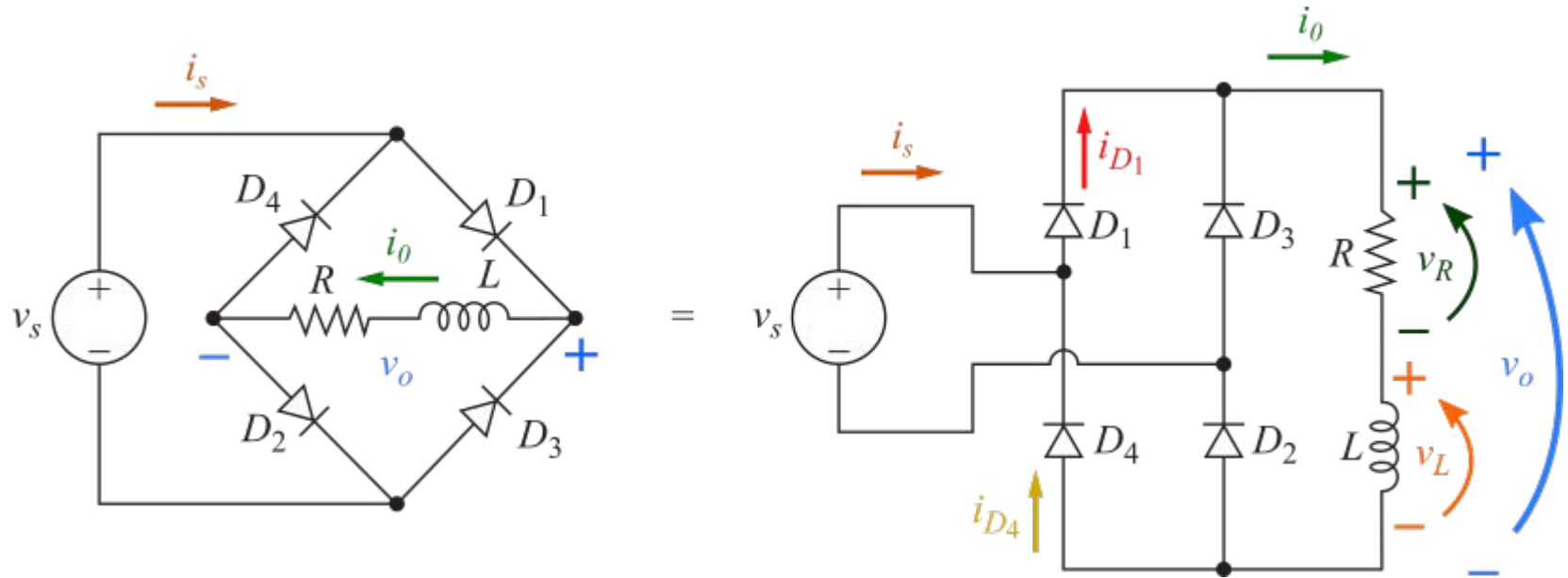


→ v_o es función de β !



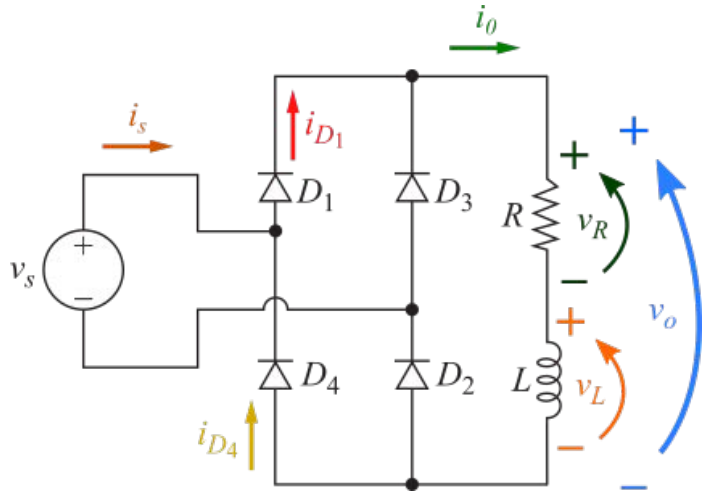
Rectificador de onda completa

- Carga resistiva-inductiva



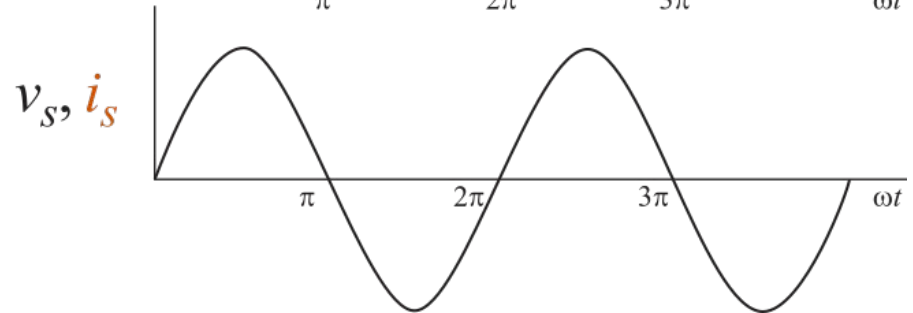
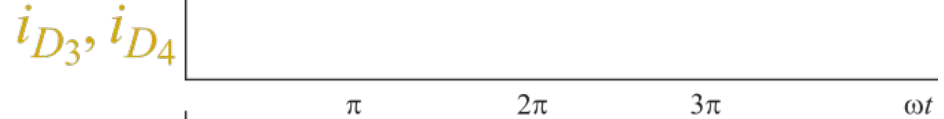
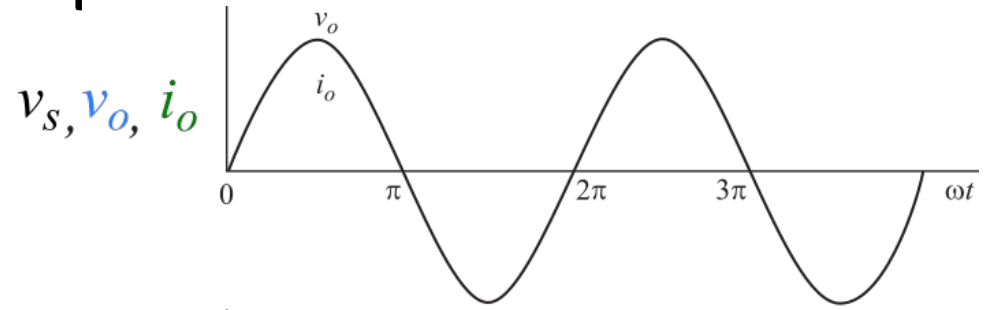
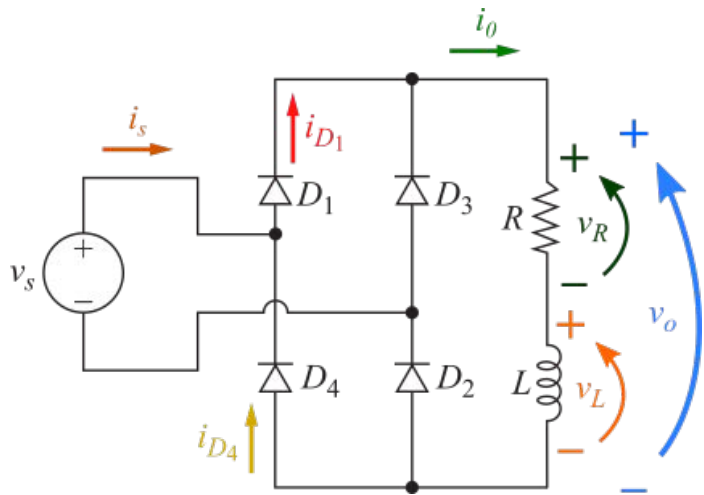
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- Carga resistiva-inductiva



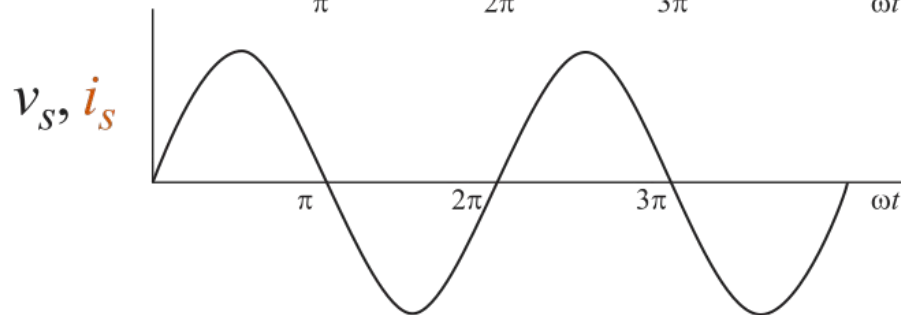
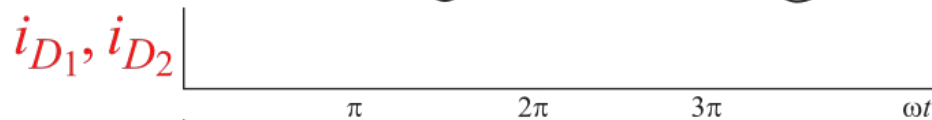
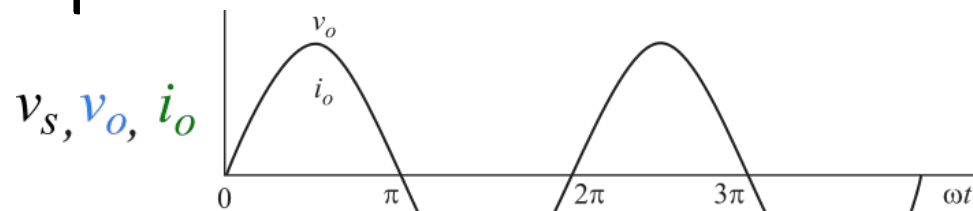
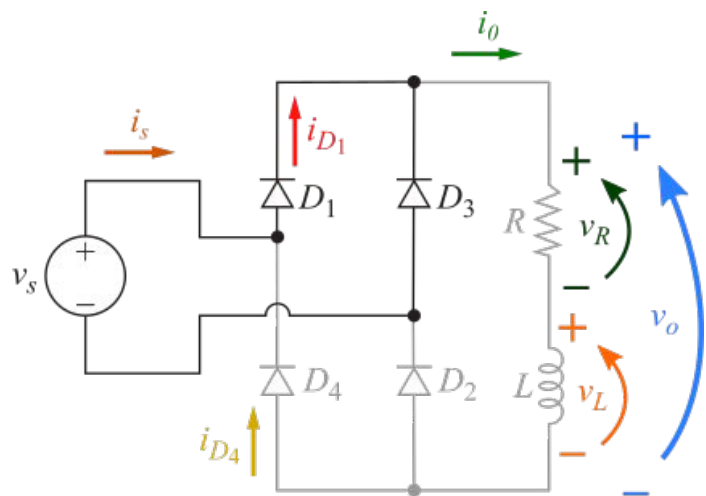
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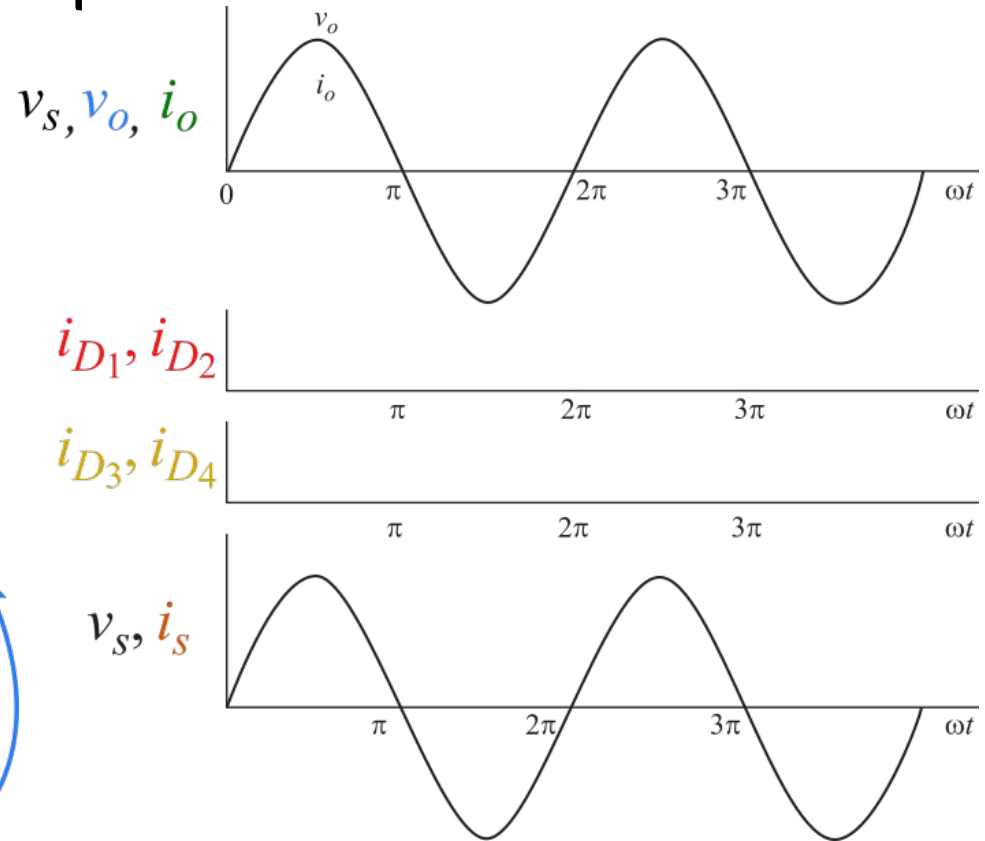
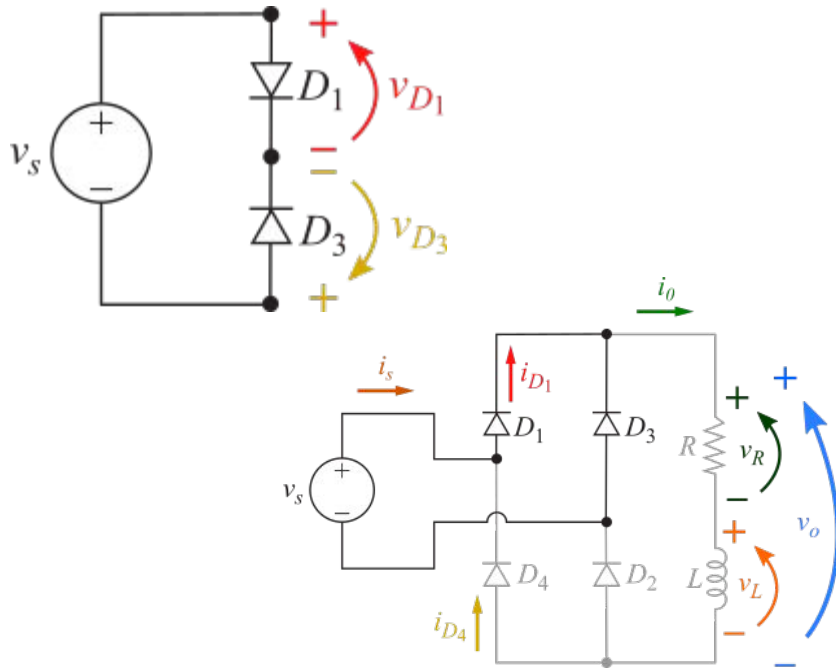
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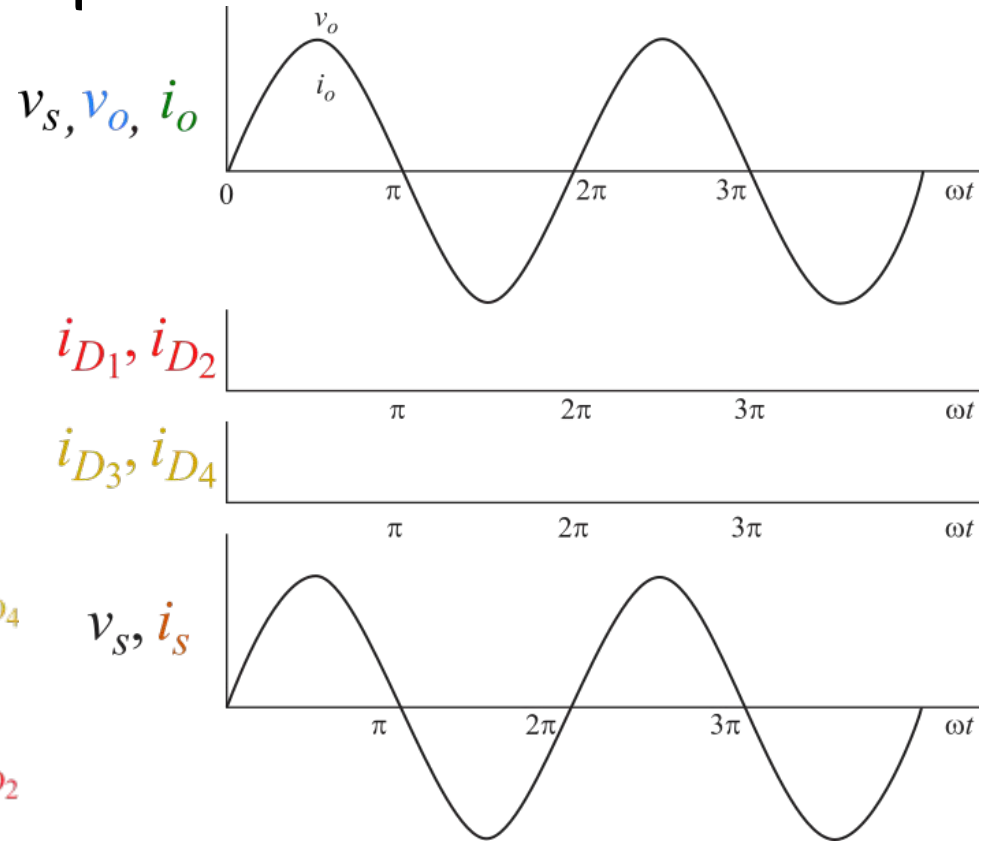
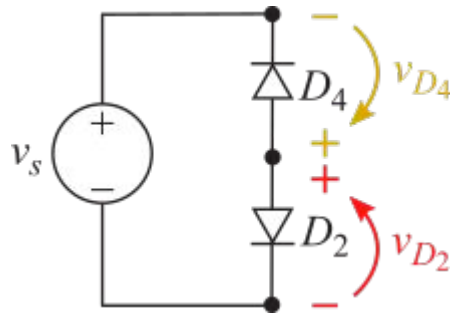
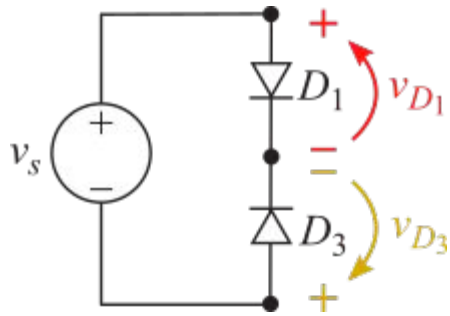
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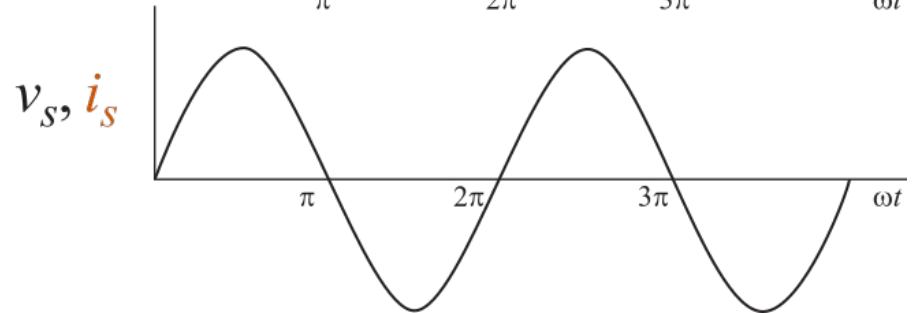
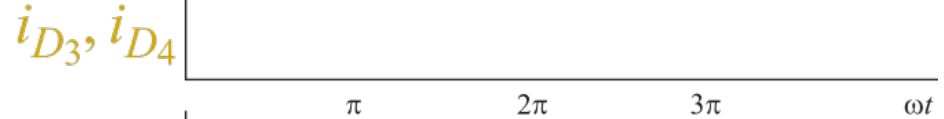
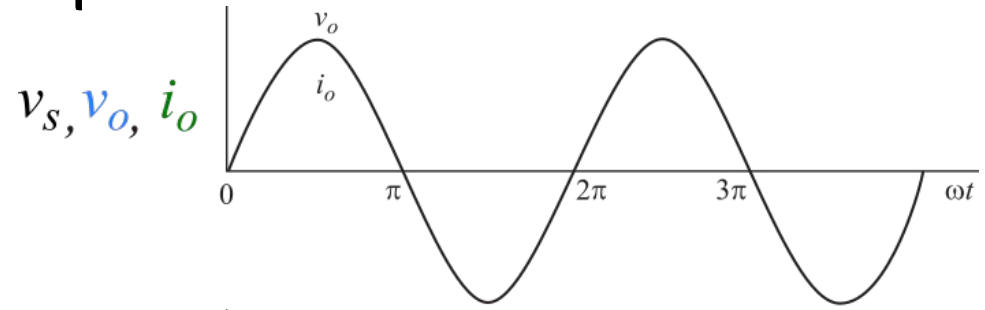
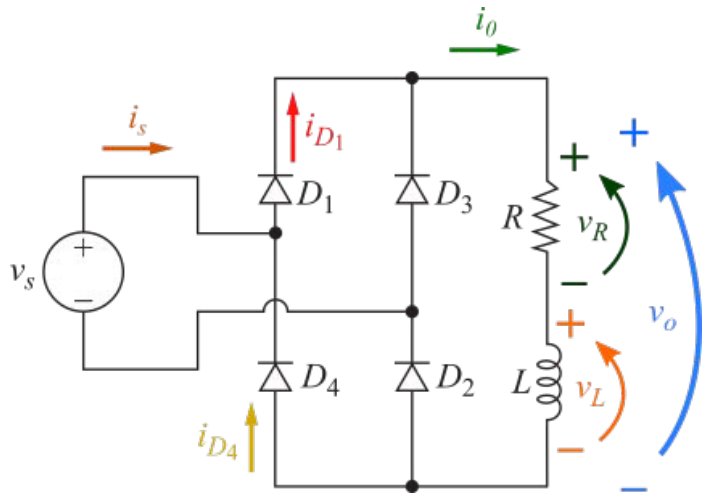
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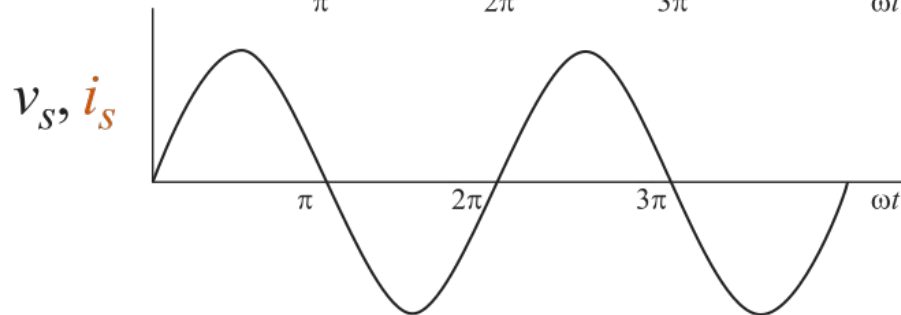
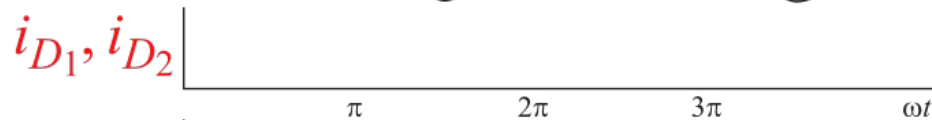
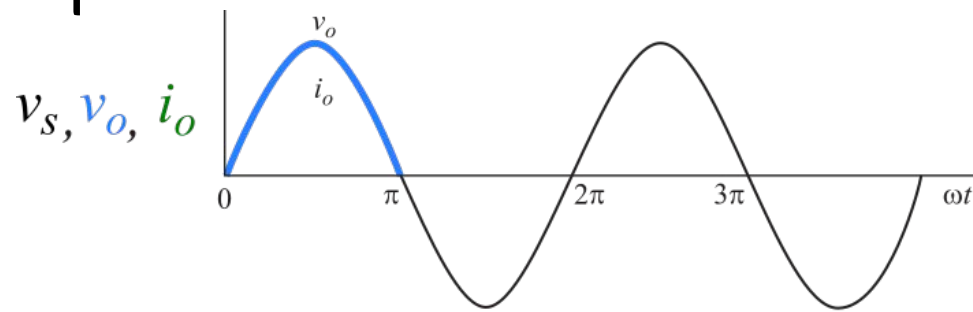
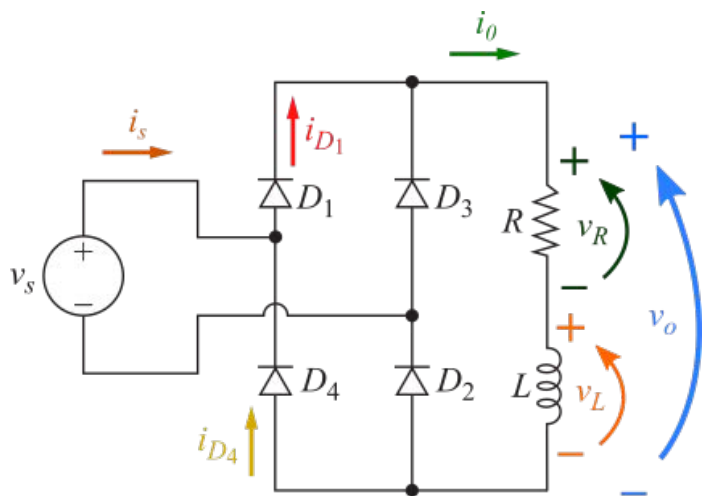
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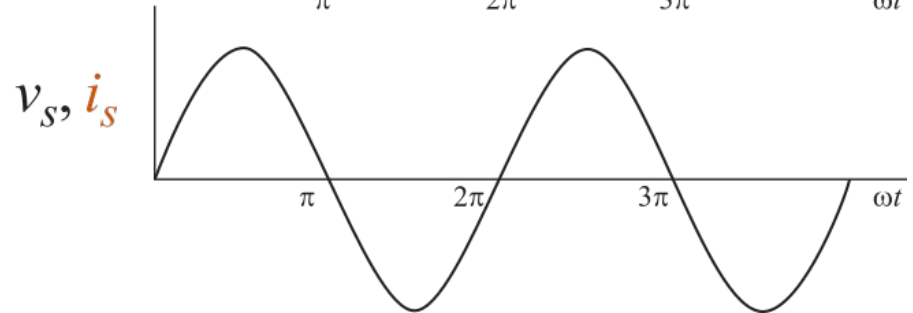
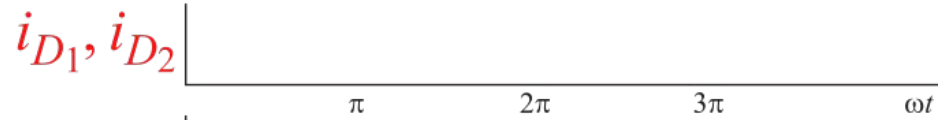
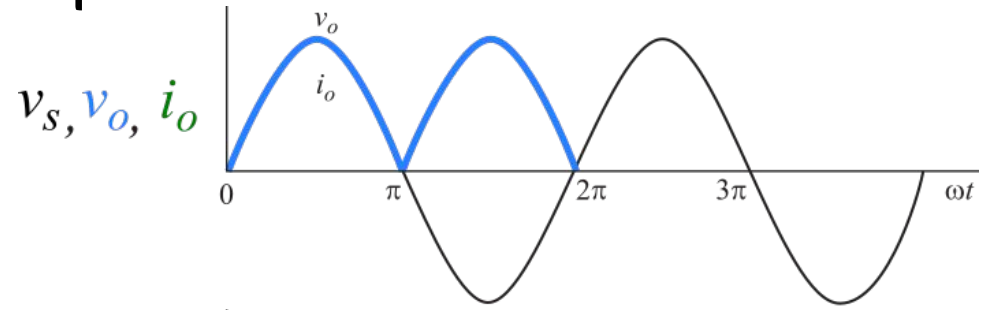
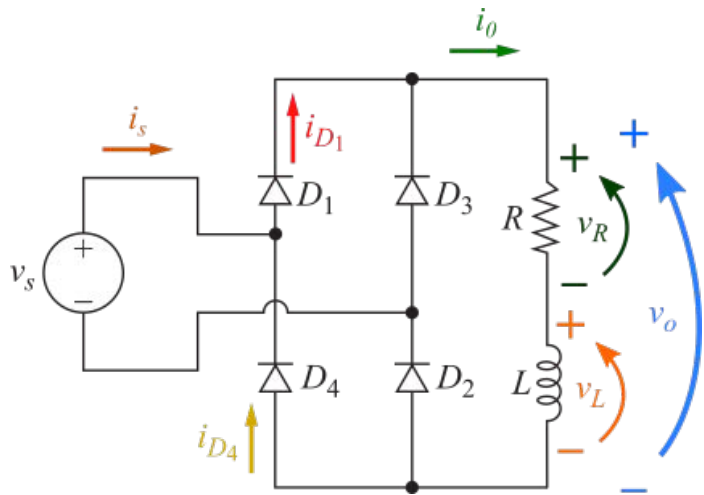
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- Carga resistiva-inductiva



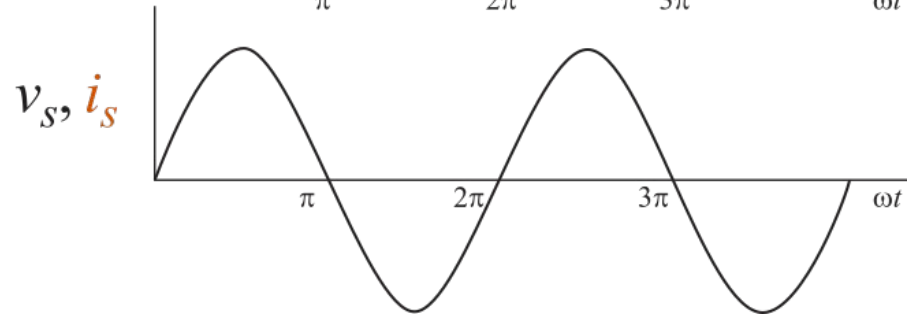
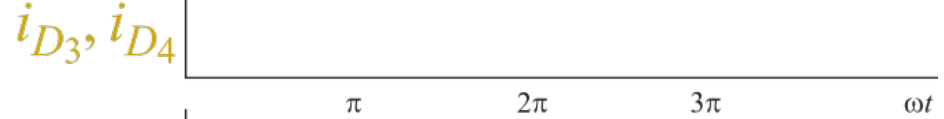
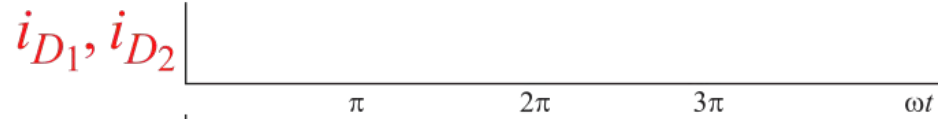
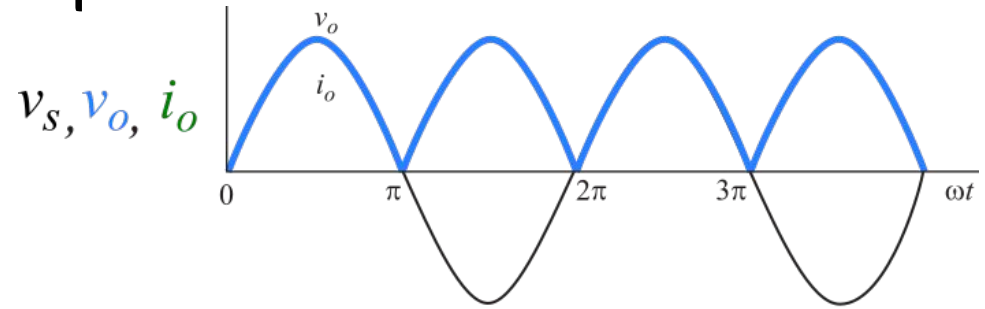
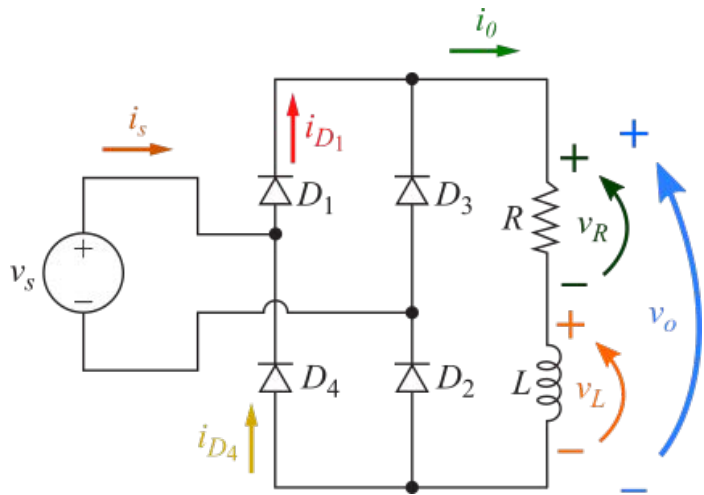
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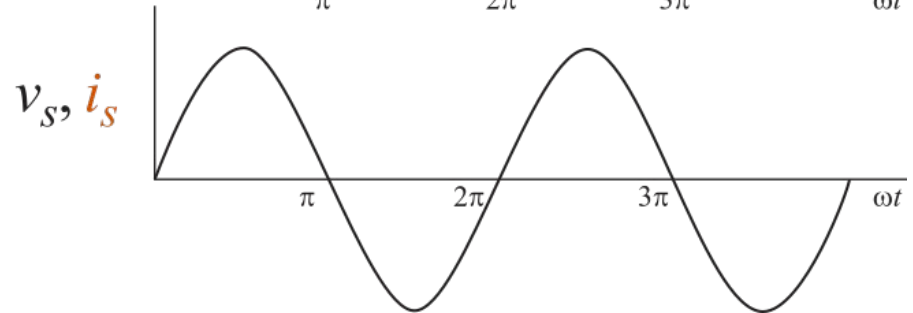
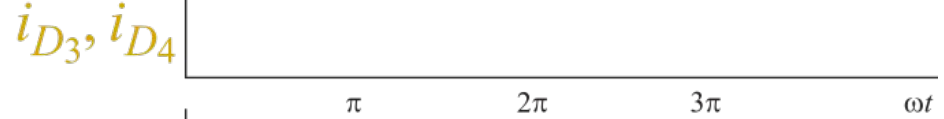
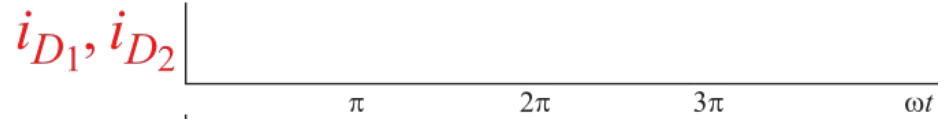
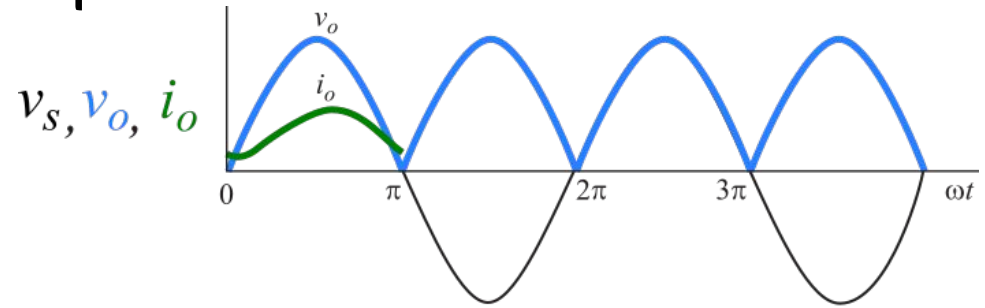
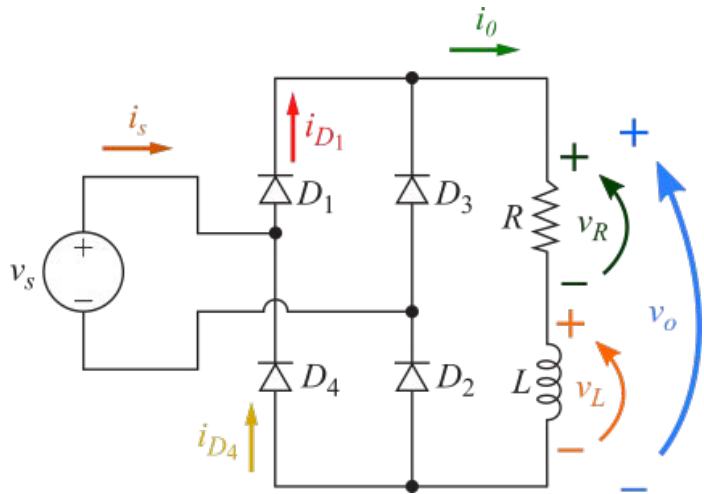
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- Carga resistiva-inductiva



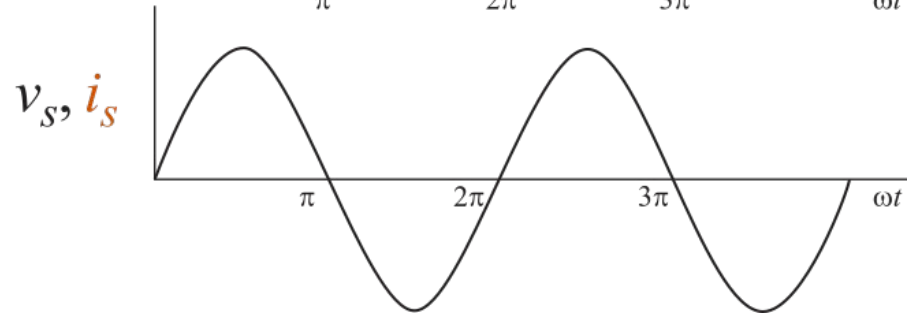
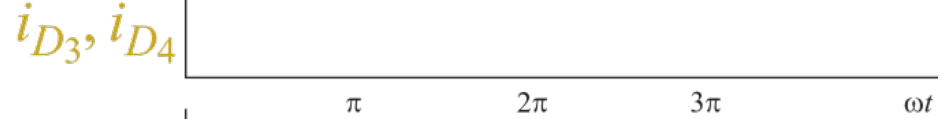
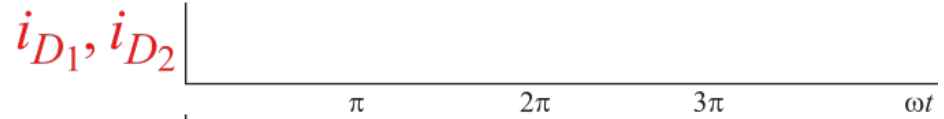
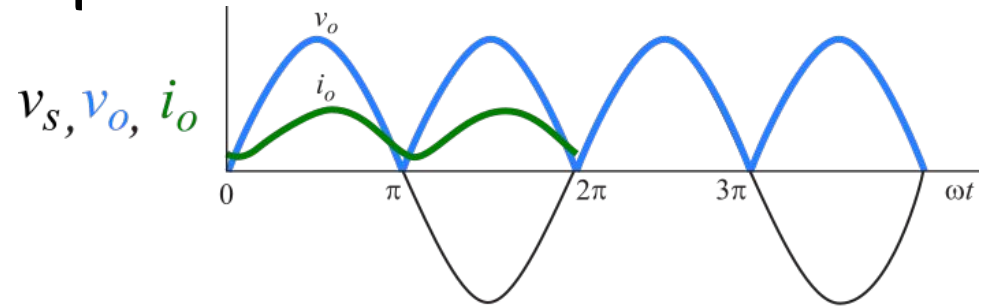
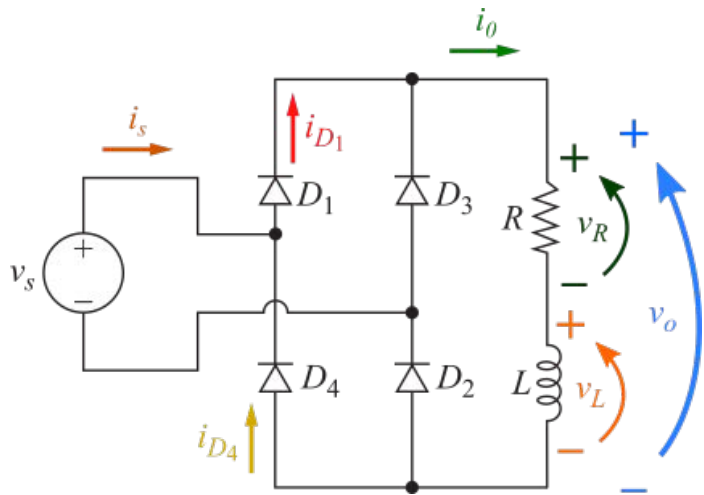
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- Carga resistiva-inductiva



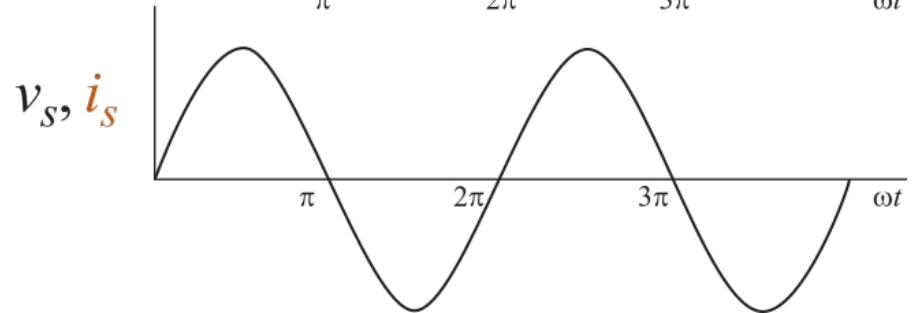
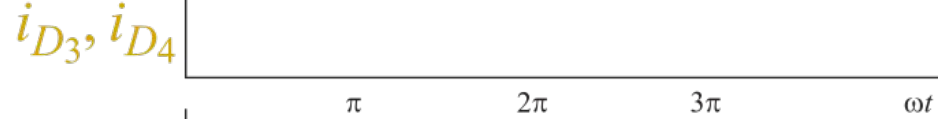
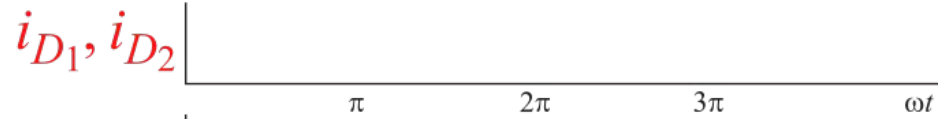
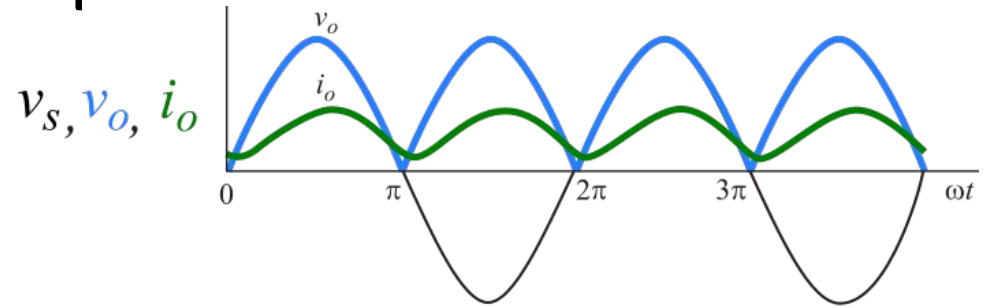
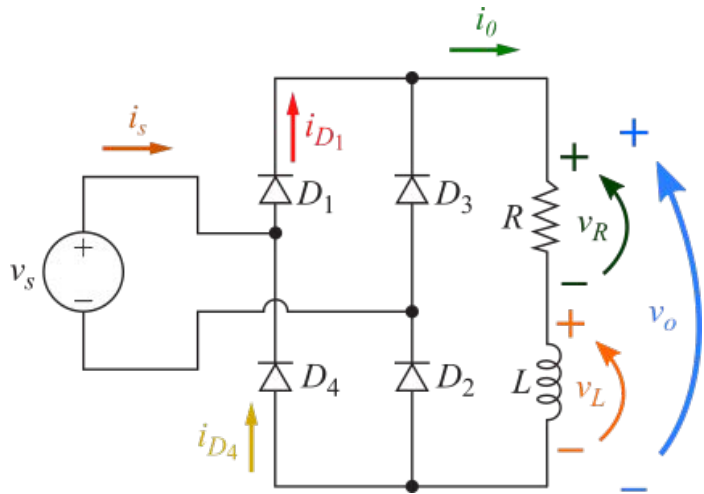
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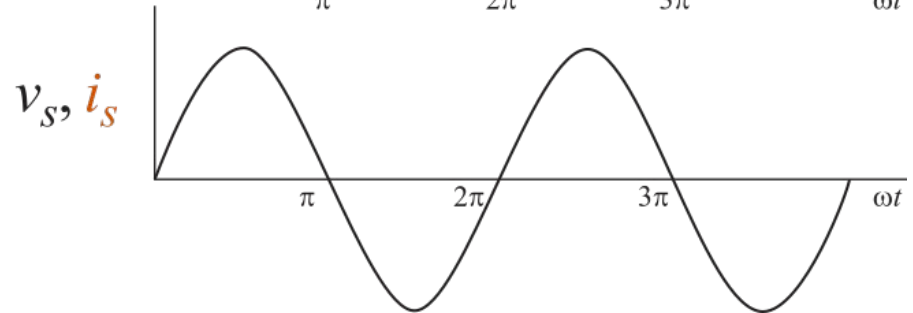
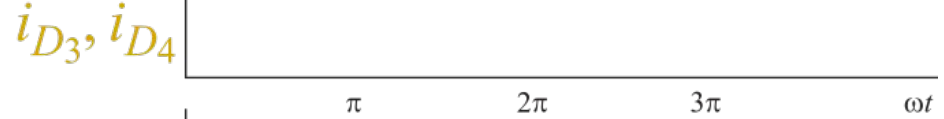
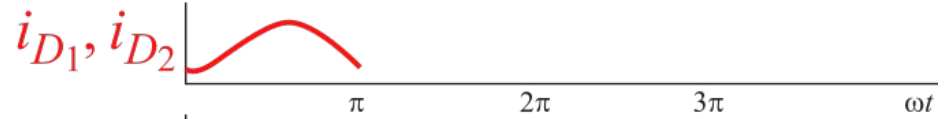
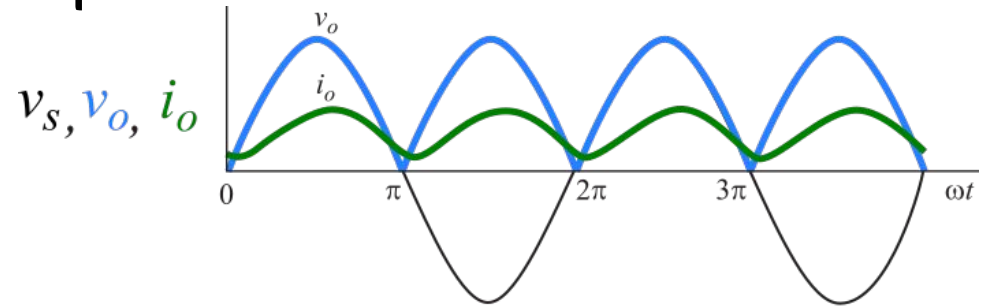
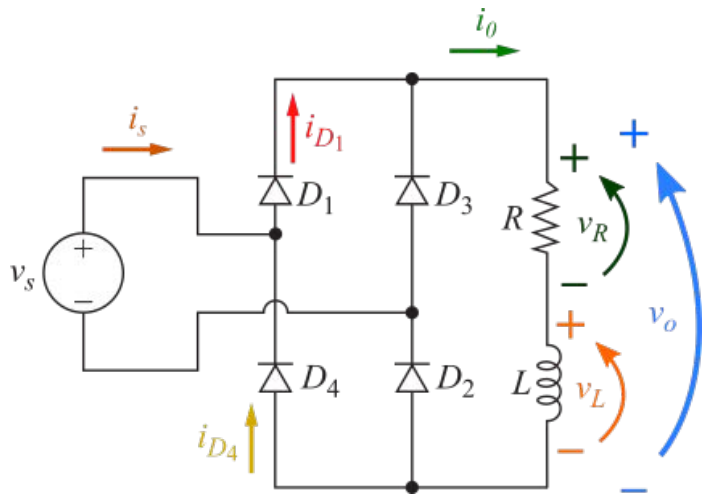
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- Carga resistiva-inductiva



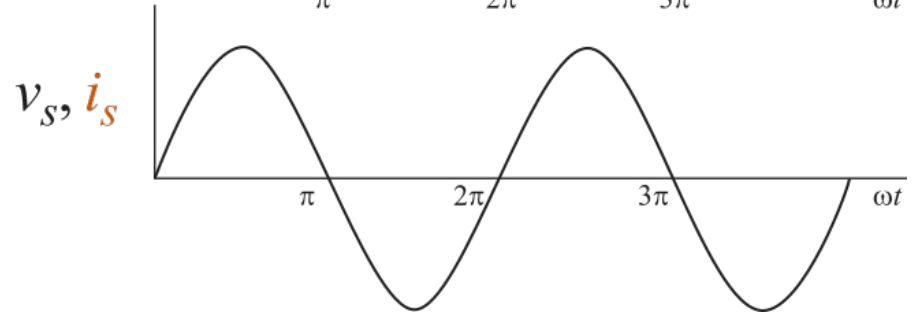
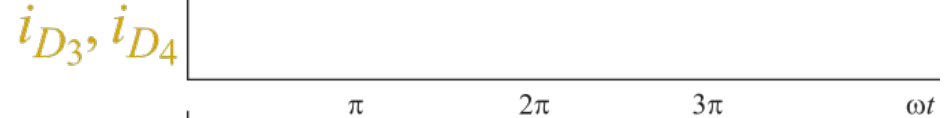
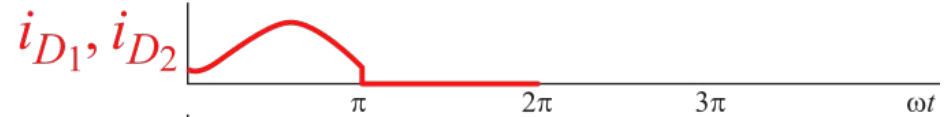
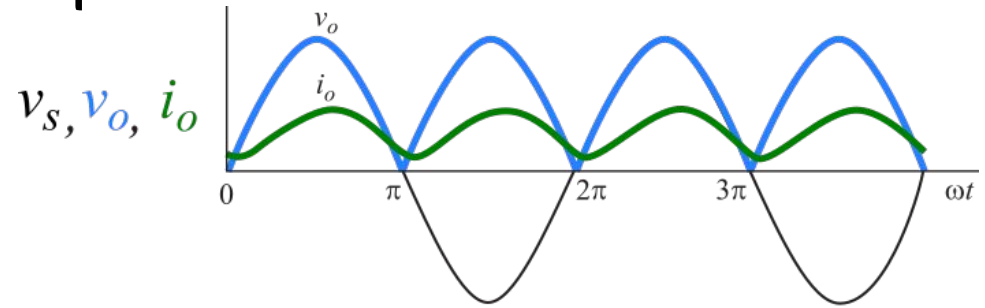
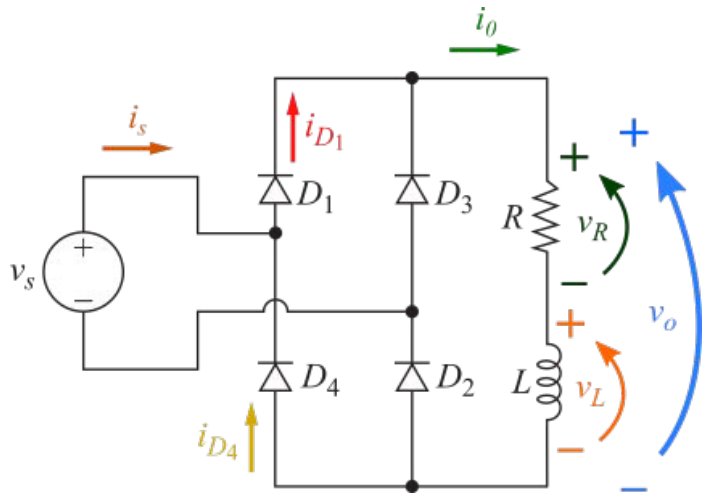
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- Carga resistiva-inductiva



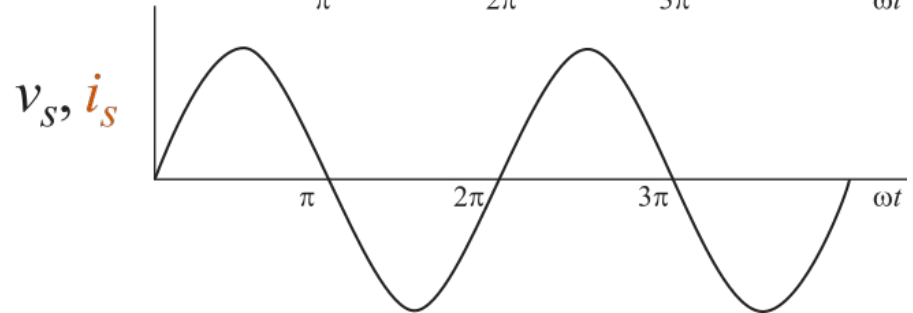
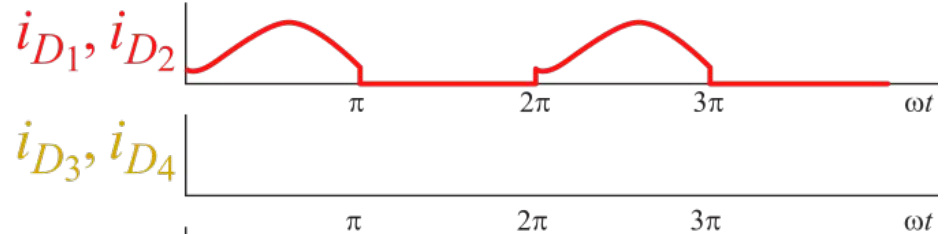
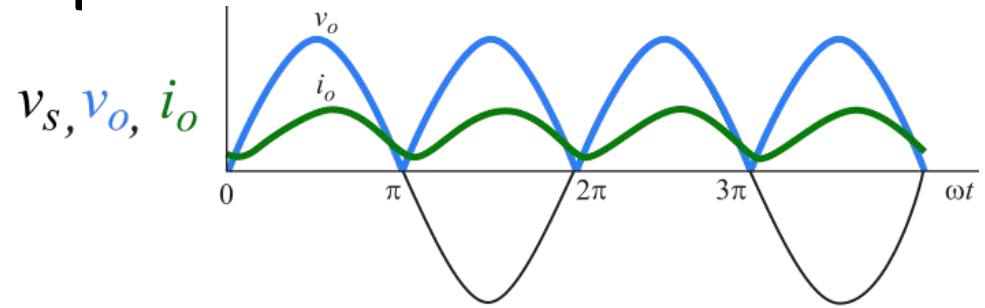
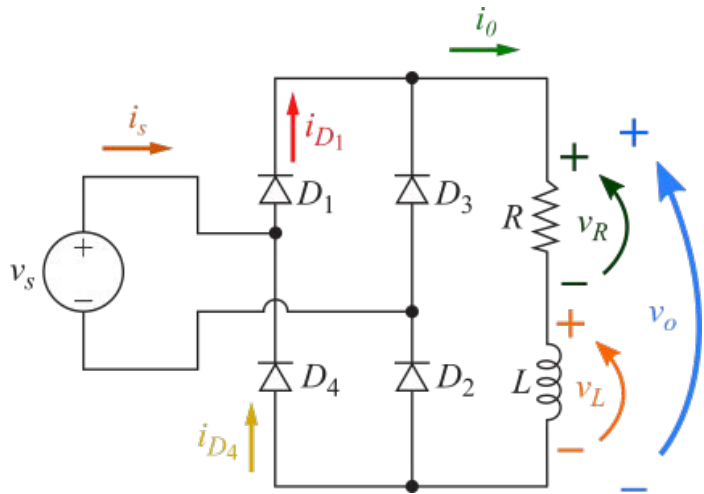
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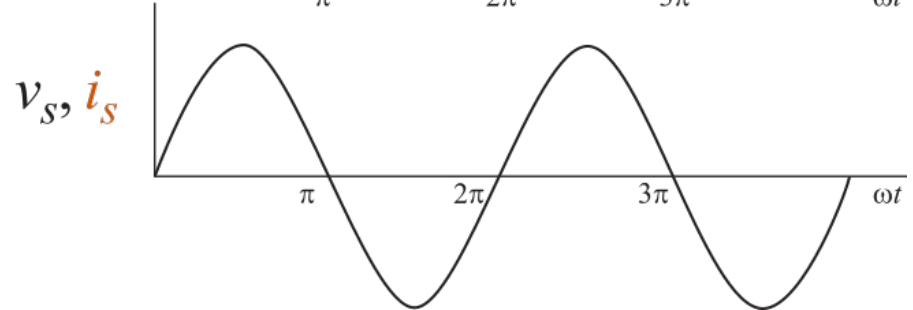
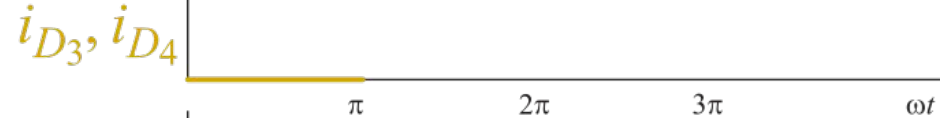
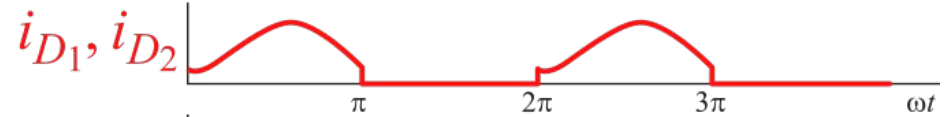
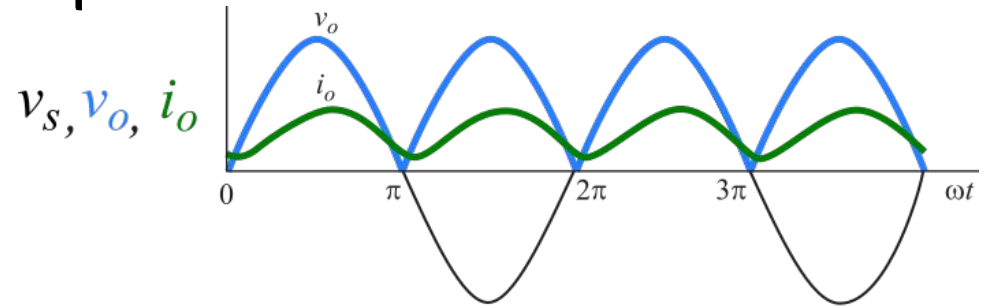
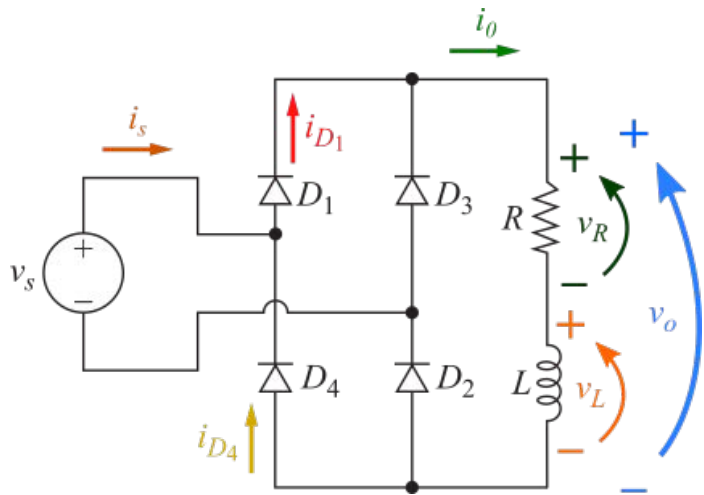
Rectificador de onda completa

- Carga resistiva-inductiva



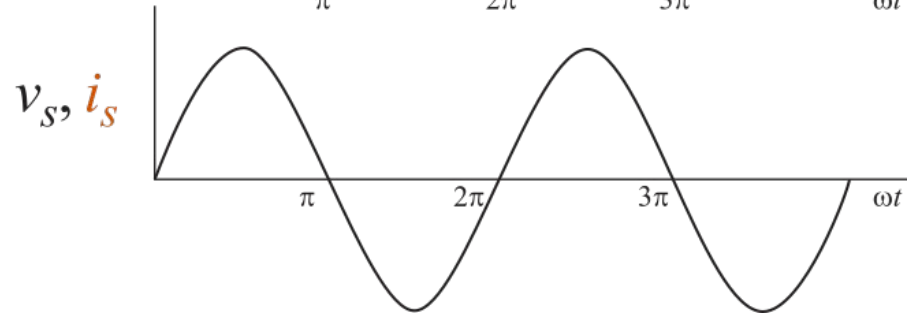
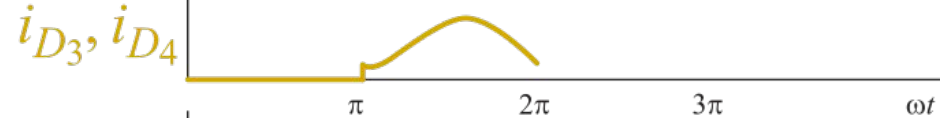
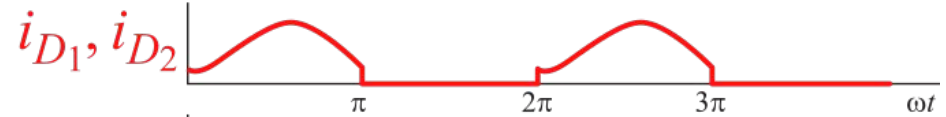
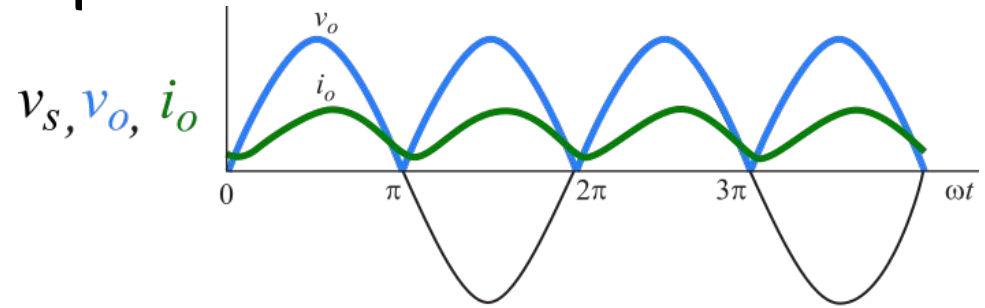
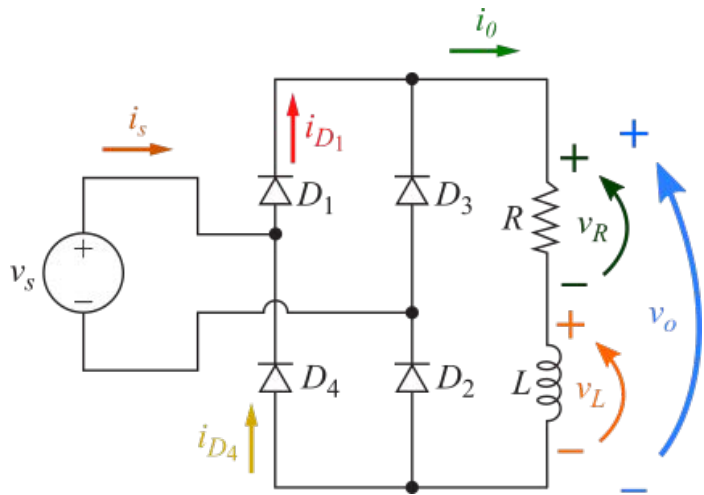
Rectificador de onda completa

- Carga resistiva-inductiva



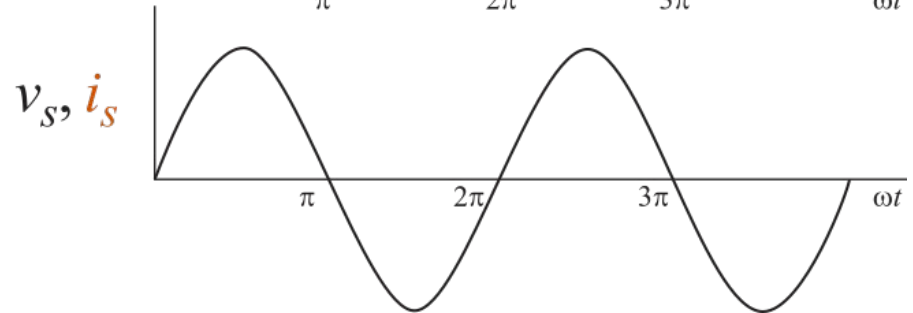
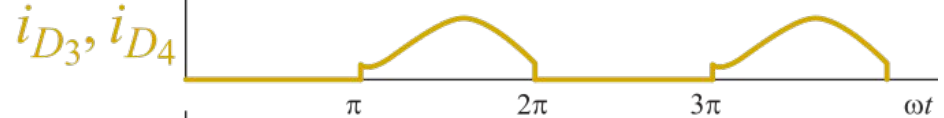
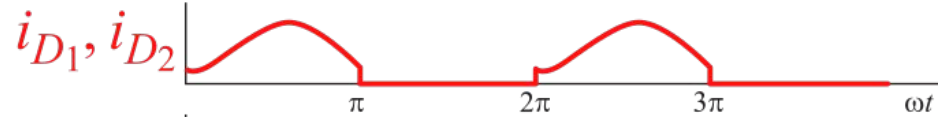
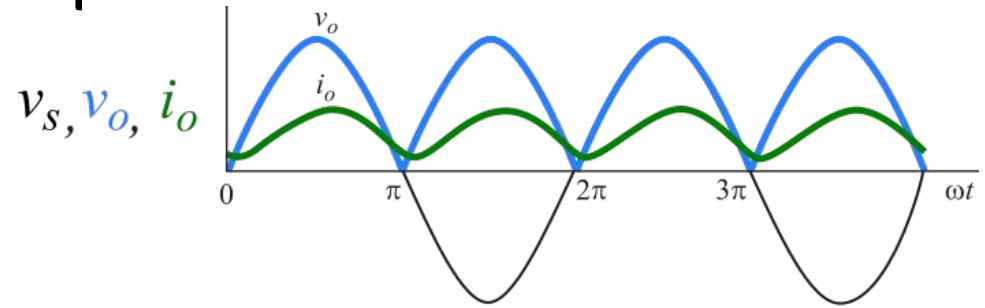
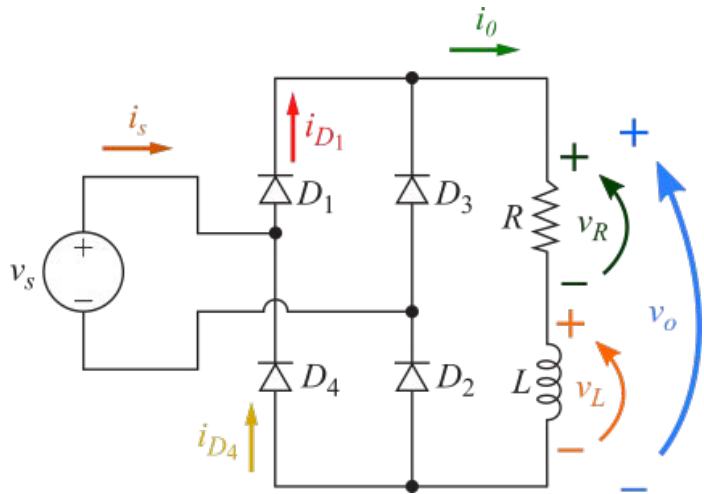
Rectificador de onda completa

- Carga resistiva-inductiva



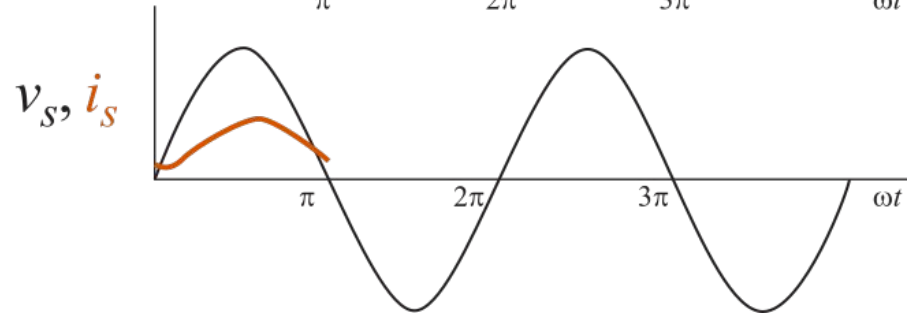
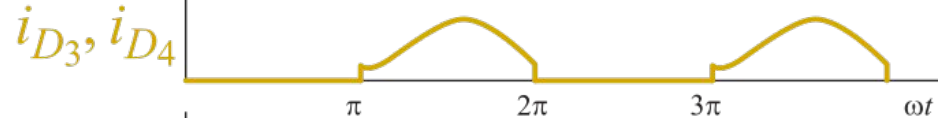
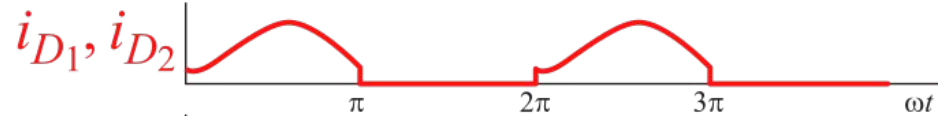
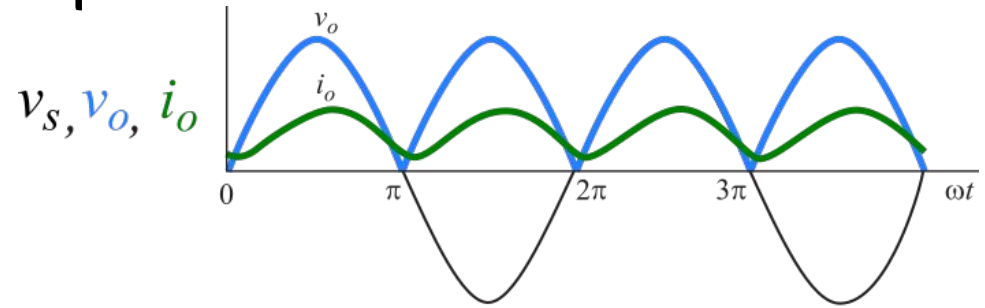
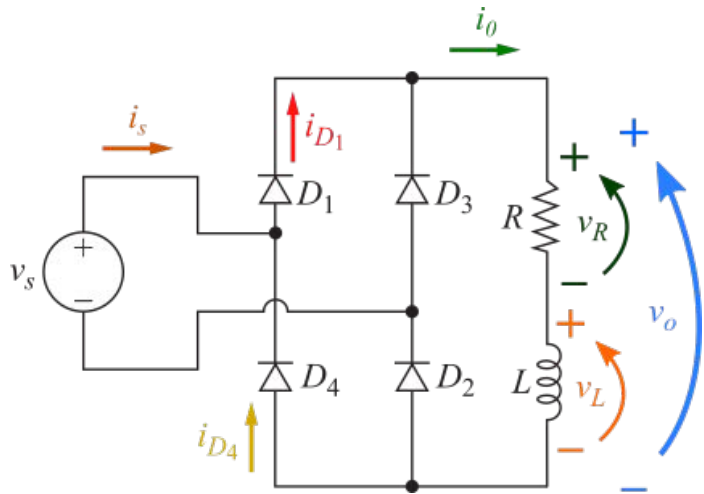
Rectificador de onda completa

- Carga resistiva-inductiva



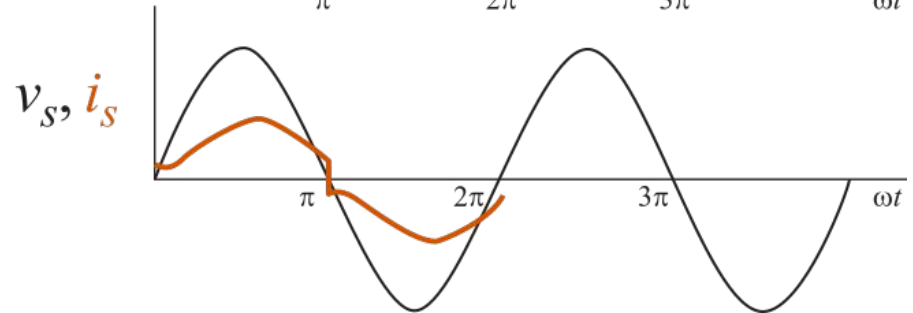
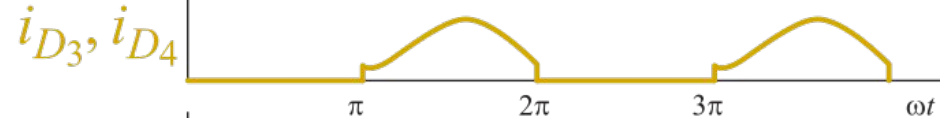
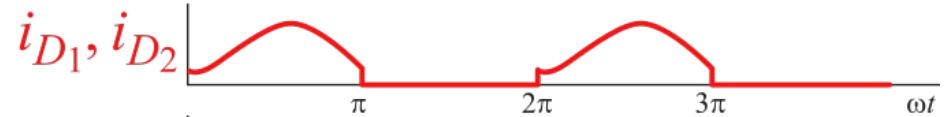
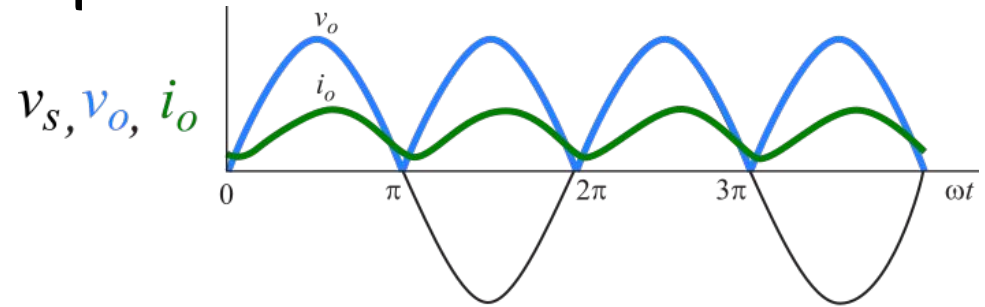
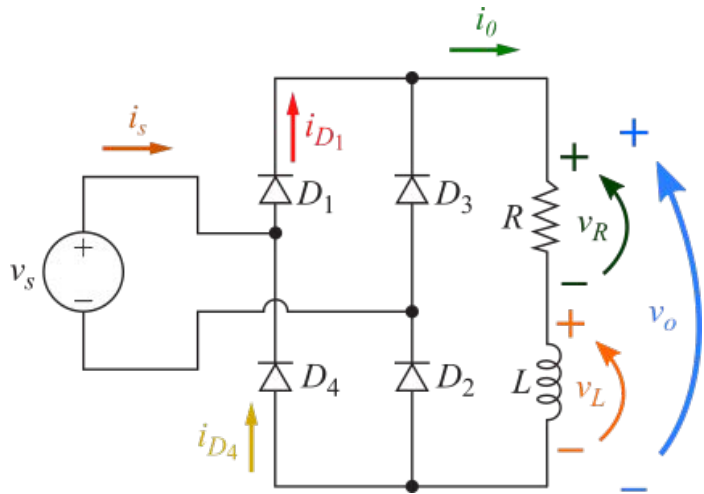
Rectificador de onda completa

- Carga resistiva-inductiva



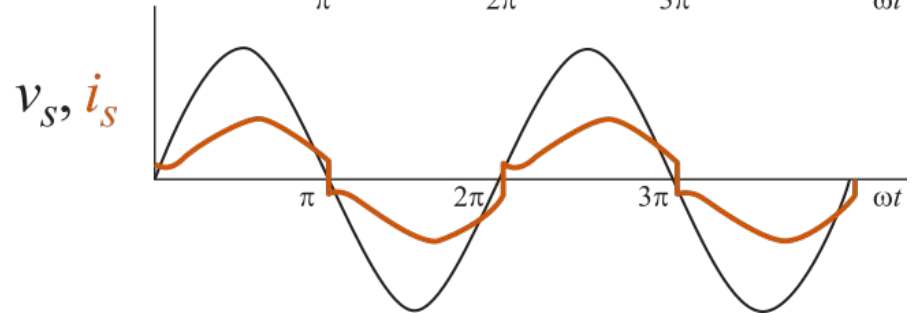
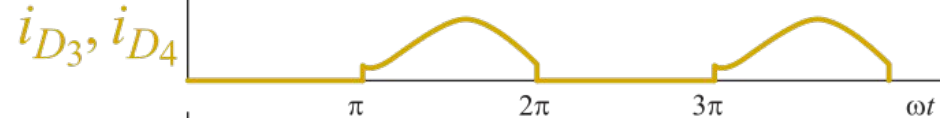
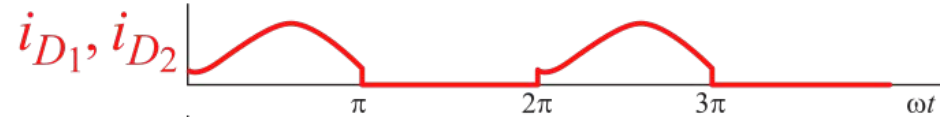
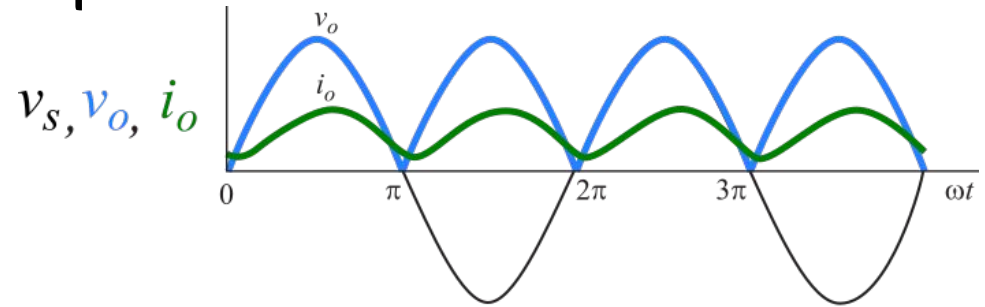
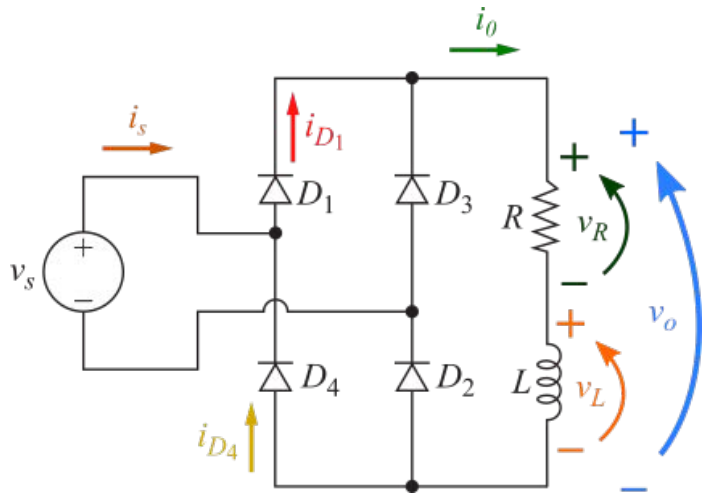
Rectificador de onda completa

- Carga resistiva-inductiva



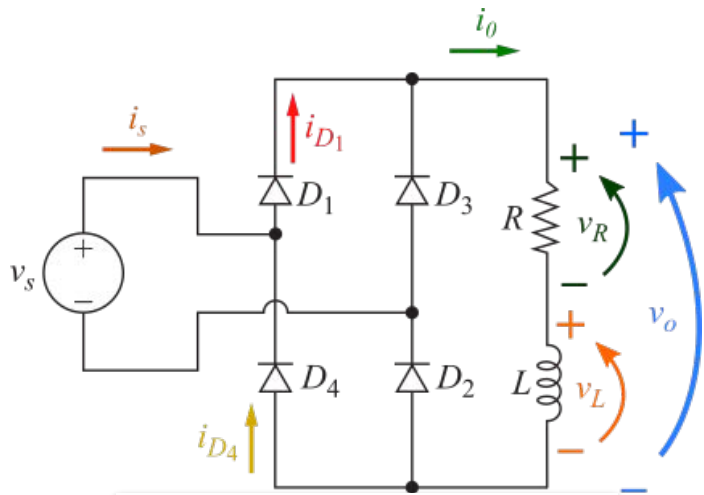
Rectificador de onda completa

- Carga resistiva-inductiva

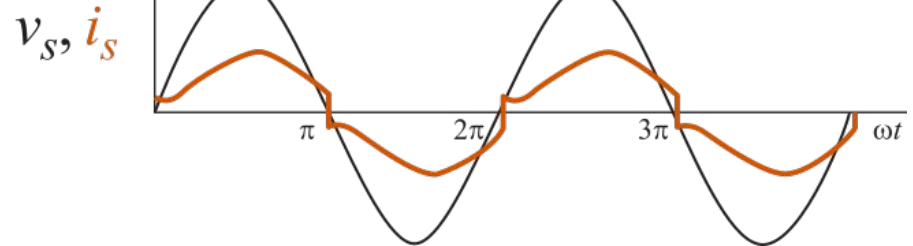
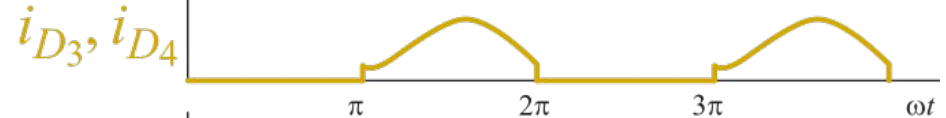
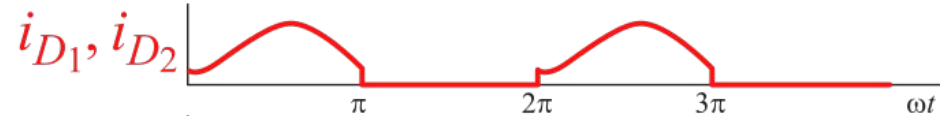
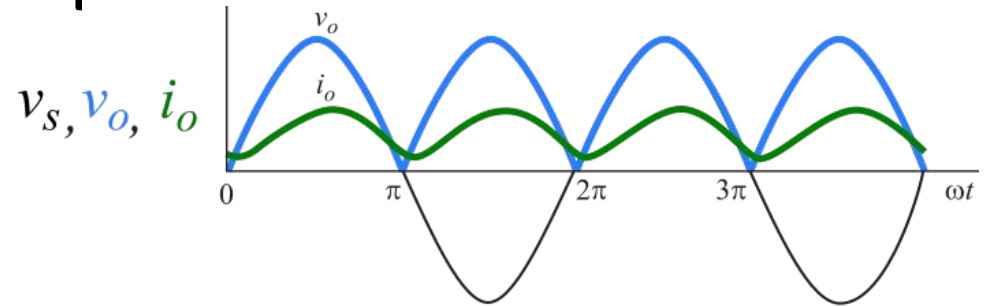


Rectificador de onda completa

- Carga resistiva-inductiva

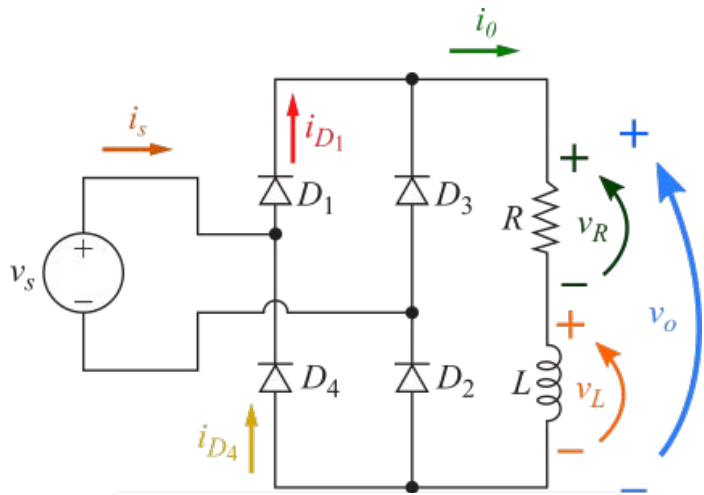


$$V_o = \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t d(\omega t) = \frac{2V_m}{\pi}$$

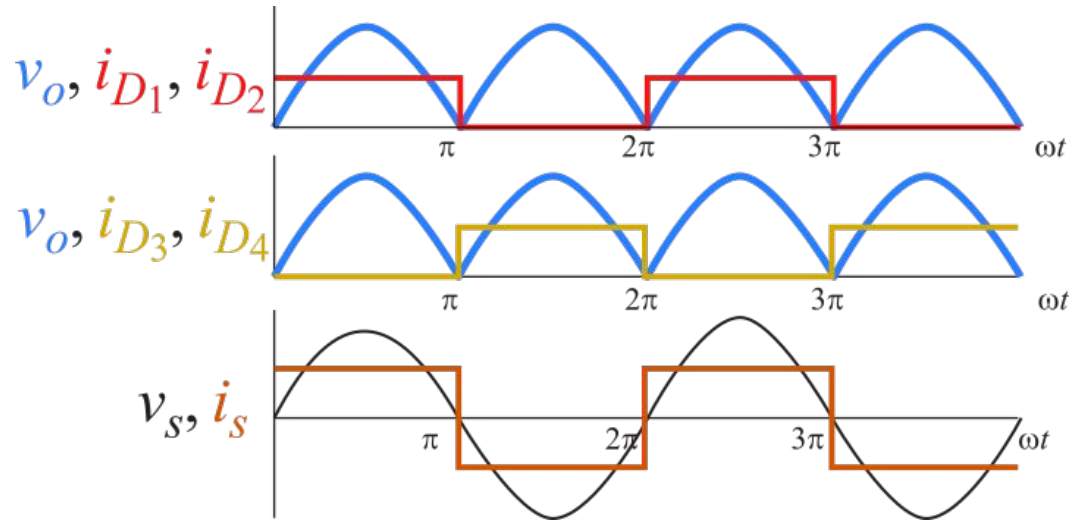


Rectificador de onda completa

- Carga resistiva-inductiva

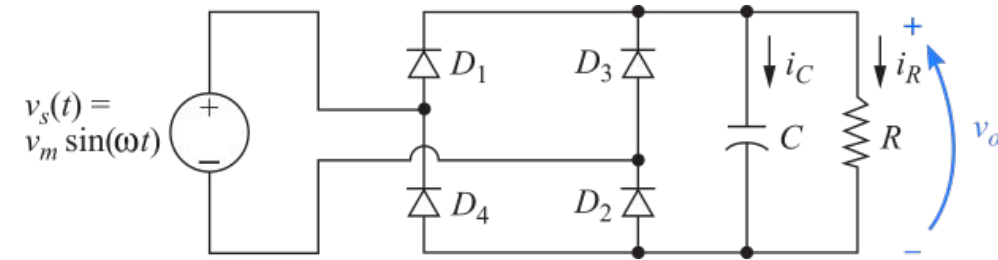


$$V_o = \frac{1}{\pi} \int_0^{\pi} V_m \sin \omega t d(\omega t) = \frac{2V_m}{\pi}$$



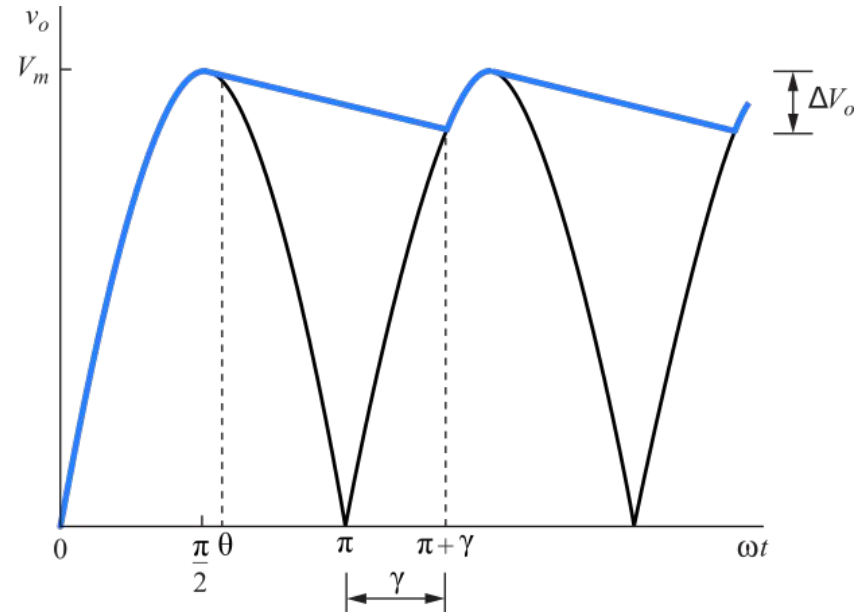
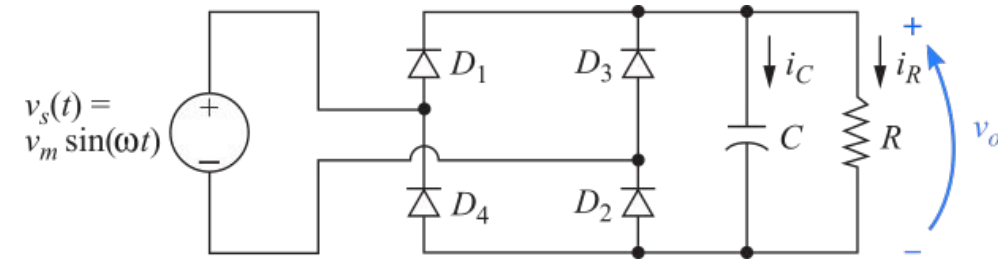
Rectificador de onda completa

- Filtro a capacitor (carga R-C)



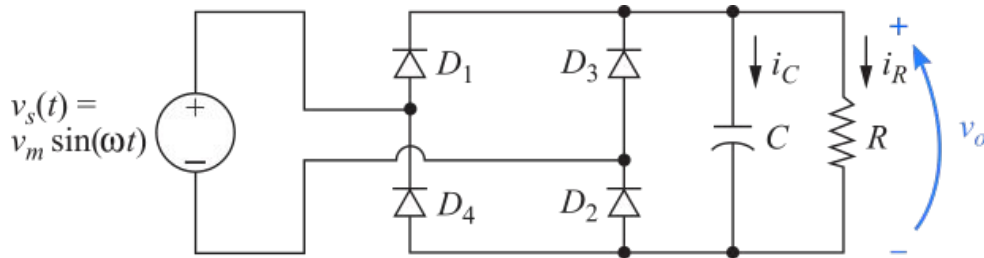
Rectificador de onda completa

- Filtro a capacitor (carga R-C)

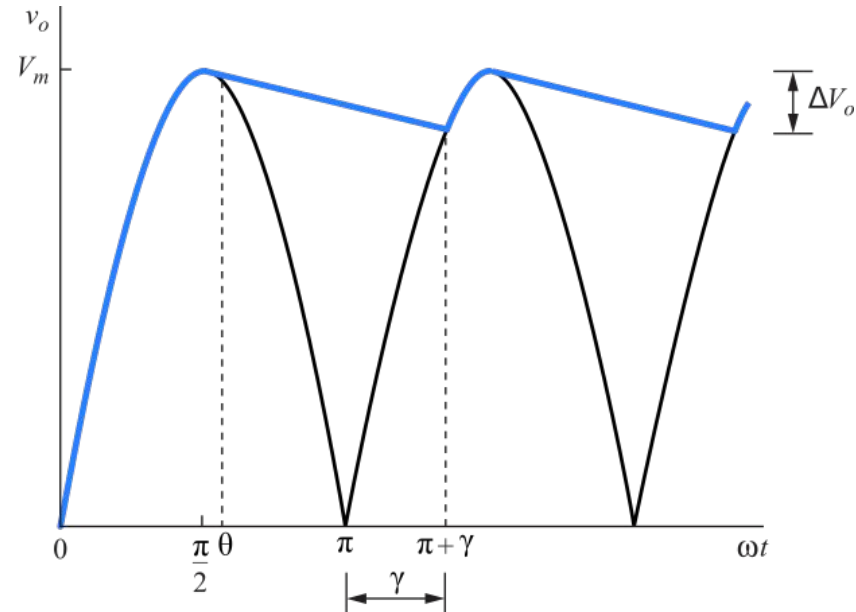


Rectificador de onda completa

- Filtro a capacitor (carga R-C)

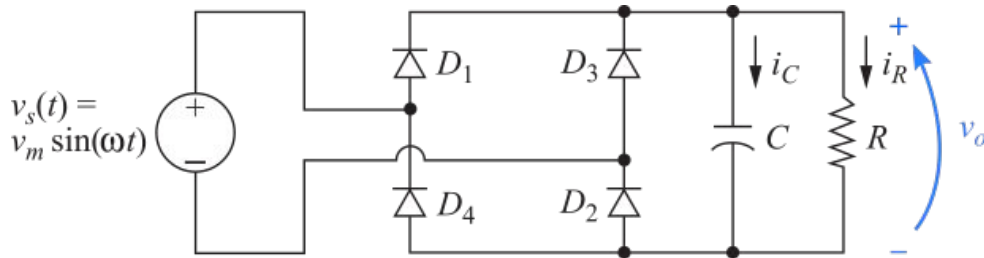


$$\Delta V_o \approx V_m(1 - e^{-\pi/\omega RC})$$



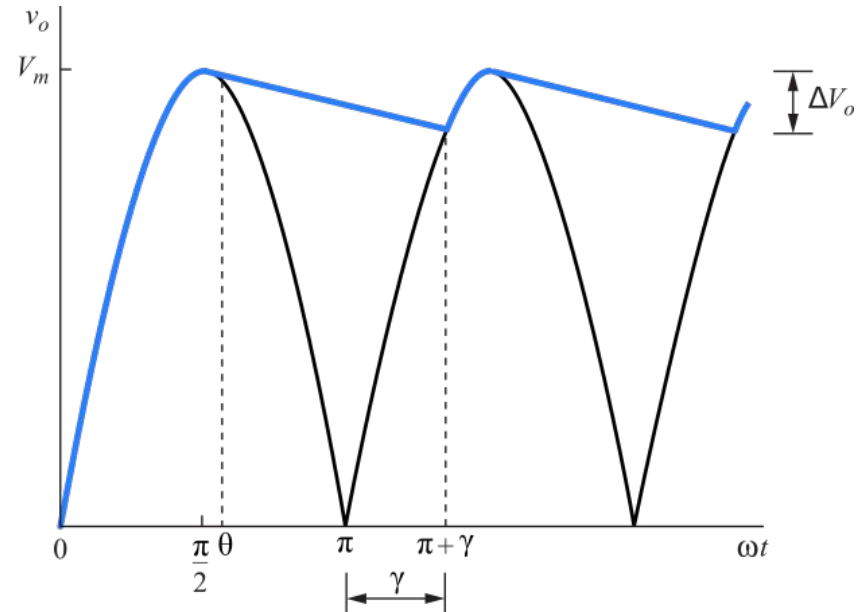
Rectificador de onda completa

- Filtro a capacitor (carga R-C)



$$\Delta V_o \approx V_m(1 - e^{-\pi/\omega RC})$$

$$\Delta V_o \approx \frac{V_m \pi}{\omega RC} = \frac{V_m}{2fRC}$$



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